

POPULATION SHOCKS, MIGRATION FLOWS, AND PERSISTENT REGIONAL DYNAMICS ^{*}

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Abstract

What are the local and aggregate effects of local population shocks, and how do they propagate through endogenous migration? To address this question, I combine a panel SVAR with an external shift-share instrument that leverages lagged birth-rate variation to estimate the response to a 1% increase of migration inflows in U.S. commuting zones. The shock generates persistent inflow dynamics, raises output per worker by about 0.2% at its peak, and increases housing prices by roughly 0.45%, with the latter remaining elevated years after the initial shock. I then develop a quantitative dynamic spatial model, and discipline it by matching the responses of gross inflows, output per worker, and housing prices, which capture migration adjustment, agglomeration, and congestion. In the model, population-stock shocks alter local attractiveness through these forces and thereby induce endogenous gross inflows and outflows. I then use the model to birth-rate changes as well as policy-driven labor removal and redistribution. Regions with higher fertility experience a rise in output per worker along with a larger increase of housing prices. On the other hand, labor removal lowers aggregate output per capita by about 0.5% in the long run, whereas redistribution preserves the long-run steady state. However, depending on the skill composition of relocated workers, the latter policy generates temporary effects ranging from a gain of about 0.7% to a loss of about 1.3%, with convergence taking more than 50 years..

Keywords: Migration, Labor Mobility, Economic Growth

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1 Introduction

The effects following local population shocks, such as evolving demographics and sudden deportations, are not exclusively determined by the initial change in size. The gain or loss of workers impacts productivity, congestion and ultimately affordability of affected areas. Housing prices can rise quickly while agglomeration forces that increase productivity may accrue only gradually. As these forces evolve, they alter local attractiveness and induce dynamic adjustments in gross inflows and outflows over time, making migration both a source of population change and an endogenous margin of adjustment. The resulting transition determines whether local effects dissipate, persist, or spread across space, and therefore how geographically concentrated shocks can have aggregate consequences. Yet causal studies of migration generally identify local effects without a general-equilibrium transition, while quantitative spatial models often discipline adjustment with steady-state or cross-sectional moments.

This paper connects the two: I estimate the dynamic causal effect of migration inflows and use these estimates to quantify how demographic and policy-driven population-stock shocks propagate through the U.S. economy. I proceed in three steps. First, I estimate a panel SVAR-IV for U.S. commuting zones (CZ) and trace the joint response of gross migration flows, output per worker, wages, employment, establishment dynamics, and housing prices to an exogenous inflow shock. Second, I develop a dynamic spatial model in which market size affects firm variety and future entry costs, while fixed land and congested amenities raise local costs. I estimate the model by matching three core impulse responses—gross inflows, output per worker, and housing prices—which discipline migration adjustment, agglomeration-driven productivity, and congestion. Third, I use the estimated model to study both gradual demographic shocks, through post-sample regional birth rates, and concentrated policy shocks that remove or relocate workers across regions. The common object across all three steps is the sequence of endogenous migration flows generated when a population-stock shock changes local attractiveness.

To identify the local causal effect of migration inflows, I estimate a panel SVAR-IV model using shift-share instruments derived from lagged birth rates. This vector autoregression structure enables me to capture the intertemporal relationships between population inflows and other key variables. As a result, I can account not only for the direct impact of migration inflows on economic outcomes but also for the potential feedback effect, where favorable economic conditions attract additional future migration. To address potential simultaneity bias arising from contemporaneous productivity shocks that could drive migration into the area, I employ an instrumental variable approach. After estimating the reduced-form panel VAR, I apply the methodology of [Stock and Watson \(2018\)](#) to identify the impact of structural inflow shocks using an external instrument. Specifically, for each CZ, I predict population inflows using a shift-share instrument: the ‘pull’ factor are lagged migration

shares at the destination and the ‘push’ factor is the lagged birth rate of each origin, following a similar approach to [Karahan, Pugsley and Şahin \(2019\)](#).

The exclusion restriction of this instrument relies on the assumption that birth rates in the areas sending migrants are uncorrelated with destination-specific shocks 25 years later. Following the strategy used in the immigration literature, such as in [Altonji and Card \(1991\)](#) and [Card \(2001\)](#), the weights of the shift-share instrument correspond to the lagged share of migrants originating from each area relative to destination population. However, unlike most studies in this field, I use migration patterns from the previous year rather than a fixed pre-period. Although these shares may be endogenous, I follow [Borusyak, Hull and Jaravel \(2022\)](#) and estimate the IV at the ‘shock level’—in this case, the level of the origin-CZ sending migrants. This approach allows me to leverage the exogeneity of past birth rates while constructing a stronger instrument with inflow shares that more accurately predict current migration inflows. Additionally, it enables me to calculate the appropriate standard errors for inference, thereby avoiding the limitations of conventional shift-share methods discussed by [Adao, Kolesár and Morales \(2019\)](#).

At the CZ level, a 1% increase of migration inflows leads to persistently higher future immigration, raises wages and output, drives net establishment growth, and increases housing prices. The shock produces an immediate effect on all variables, and several responses build over the next few years. Real GDP per worker peaks near 0.20%, real average wages near 0.10%, and the employment-to-population ratio near 0.15%; all three remain positive at the end of the ten-year horizon. At the same time, some workers begin to leave the area, as out-migration settle near 0.2% above its pre-shock level by year ten. This is consistent with higher housing prices that respond more strongly than productivity or wages, rising from 0.21% on impact to about 0.46% in year three and remaining near 0.39% in year ten. The joint increase in productivity, housing costs, and both gross flows is consistent with agglomeration gains operating alongside congestion and spatial sorting.

I also provide additional evidence to shed light on the mechanisms behind the baseline results. First, I study other measures of local dynamism that show the expansion is driven by new establishments and employment growth within incumbents. Indeed, establishment net entry, gross entry, and job creation rise after the shock, while job destruction falls and establishments per worker decline. Second, I split CZs by population density (population per square mile) and re-estimate the SVAR-IV separately for each group. Output-per-worker gains are largest in smaller and less-dense CZs, whereas housing-price pressure is strongest in larger and denser markets. These estimates suggest that agglomeration externalities are greatest where initial scale is lower, while congestion spillovers dominate in densely populated locations. Third, to determine if the modest rise in average wages is mainly driven by composition effects, I examine the wage response within narrowly defined industries. I find the response is also positive across the selected industries, reducing the concern that

the aggregate wage response is driven only by shifts across broad sectors.

These results underscore why it is essential to estimate the dynamic effects of migration inflow shocks. First, the impulse responses show that the impact is not concentrated on arrival: several outcomes continue to build or decline over subsequent years, reflecting slower adjustment mechanisms such as housing construction, changes in firm entry, and additional migration inflows (and outflows) in subsequent years. Second, estimating these responses to flow shocks helps discipline the transition following shocks to local stocks. When policy or demographic forces change the number of residents in a place, they also change congestion and productivity. This in turn makes locations more or less attractive for households, producing a sequence of inflows and outflows that update local population stocks and economic outcomes. The IRFs identify the local propagation of this sequence.

We need a quantitative model to map the local responses into aggregate transition paths, and incorporate the spatial general-equilibrium interactions. I therefore develop a continuous time spatial economic model in which forward looking heterogeneous workers choose where to live each period, subject to mobility costs on the intensive and extensive margin. Workers differ in raw skill, and the return to skill varies across regions. Furthermore, the model includes static agglomeration externalities à la [Romer \(1990\)](#) as well as local dynamic spillovers similar to the ones in [Peters \(2022\)](#), where a larger local mass of firms in a region decreases the cost of entry in the future. On the other hand, endogenously congested amenities as well as a housing sector with fixed land make locations less attractive as population increases. Mobility frictions slow the spatial redistribution of labor and, when combined with intertemporal productivity spillovers, allow migration shocks to have persistent and time-varying effects on local outcomes. This is what we observe in the empirical IRFs.

To discipline the main parameters of the model I use impulse response matching. In particular, I simulate the transition path of the economy implied by the empirical birth rates in the shift-share instrument. I then replicate the estimation described above using the simulated data to obtain model generated IRFs. The parameter estimates I obtain are the ones that minimize the distance between the simulated and empirical responses of gross inflows, output per worker, and housing prices. These three outcomes discipline, respectively, endogenous migration adjustment, agglomeration-driven productivity, and housing-related congestion. Other parameters are estimated using calibration by targeting cross-sectional evidence with moments generated in the initial steady-state.

I use the estimated model to study two types of population-stock shocks. The first is a gradual demographic change. In the calibration, I use birth-rates from the same time frame as the SVAR-IV estimation. For this exercise, I feed the birth rates corresponding to the 25 years after the estimation period into the model. On average this represents a small permanent drop of 0.02 percentage points in fertility. However, this hides some heterogeneity as some regions experience a rise whereas in others birth rates fall. The

results show how output per worker increases by 0.25% in the former, along with varieties and population. However, these same regions also see real incomes continuously fall by 0.3% several years after the shock as housing prices rise. In addition to informing current debates on the overall fall of birth rates, these counterfactuals show how the effects of demographic change can be geographically heterogeneous and take a long period to materialize.

The second type of shock is immediate: I reduce population in selected regions and then compare permanent labor removal with a relocation experiment. In the first, I use the empirical geographical distribution of undocumented migrants in the U.S. and remove that share of the workforce from the economy. It is essential to emphasize that these are region-specific shocks, with the removed share varying across regions, representing between 3% and 5% of local populations. The areas directly affected by the shock experience higher amenities, lower housing prices, and real incomes that are, on average, 7.5% higher. As a result, these locations attract workers from other regions, with increased inflows and reduced outflows. However, the overall reduction in the workforce leads to a long-term decline in the number of firms and total output by 1% and 2%, respectively, across all regions. These dynamics ultimately cause aggregate output per capita to decline to a new steady state that is 0.5% below the initial level. The magnitude of these effects depends on the skill composition of the deported workers, with potentially greater impacts if higher-skilled workers are removed.

In contrast, the second policy scenario involves a relocation shock, in which the same population is redistributed uniformly across other regions. A similar policy has been implemented recently at a local level, with some states relocating undocumented immigrants to different regions within the country (Goodman et al. (2024)). The results in this paper indicate that, if applied on a national scale, such a policy would produce no long-term effects, as the economy would eventually converge back to the original steady state through migration flows. However, in the short term, the shock would create significant impacts, including a 4% decrease in housing prices across all locations and an immediate 8% rise in real incomes in areas losing workers. Moreover, the reduction in the mass of firms by 5% in these regions would lead to a regional output decline of between 2.5% and 11%, depending on the skill composition of the relocated households. At the aggregate level, relocating low-skilled households results in a slight increase in output per capita. Conversely, redistributing high-skilled workers leads to a drop in output per worker of over 1%, with recovery taking more than 50 years. Despite the different nature of each counterfactual, all three exercises generate the same mechanism: a stock change shifts agglomeration and congestion, alters local attractiveness, and triggers endogenous gross migration flows.

Related literature This paper follows previous research on market size, labor mobility, and economic growth. From a theoretical perspective, Duranton and Puga (2004) summarize the main mechanisms driving agglomeration economies. These include dividing the

fixed cost of indivisible inputs, more efficient matching, as well as spillovers from knowledge creation, diffusion, and accumulation (Lucas, 1988; Moretti, 2004; Crews, 2023). The relevance of each mechanism, and ultimately the effect of migration inflows, depends on the time frame. For example, a fixed capital stock in the short run causes wages to drop in the immediate years following worker inflows (Borjas, 2013). By estimating a dynamic model over several years, the following analysis determines how the relative importance of each mechanism evolves.

Numerous empirical studies estimate the response to inflows and obtain ambiguous results. Using previous immigration shares as an instrument, some papers find a negative effect on native employment and wages (Altonji and Card, 1991; Card, 2001). Similarly, the wage elasticities estimated by Borjas (2003) imply immigration to the US in the 1980s and 1990s reduced wages by 3.2%. Nevertheless, as the survey by Edo (2019) points out, the estimated impact on wages, employment, and other outcomes depends on the methodology, period, and country under study. Some papers have found evidence of agglomeration economies with positive spillovers on manufacturing productivity (Greenstone, Hornbeck and Moretti, 2010; Kline and Moretti, 2014) as well as employment opportunities (Moretti, 2010). Most of these studies estimate a single elasticity for multiple periods and adopt reduced-form strategies that do not account for the intertemporal correlation between migration and other economic measures.

To incorporate these interactions, this paper models local population inflows and outflows, along with other outcome variables, using a panel VAR structure. Barcellos (2010) adopted this approach in her analysis of migration between US states whereas Boubtane, Coulibaly and Rault (2013) did so for 22 OECD countries. To identify the dynamic response to population inflow shocks, these studies rely on ordering assumptions and Cholesky decompositions of the variance-covariance matrix. I avoid making such assumptions by estimating the coefficients of interest using an IV strategy as described by Stock and Watson (2018) and applied in the fiscal multipliers (Mertens and Ravn, 2013) as well as the monetary policy literature (Gertler and Karadi, 2015).

The use of exogenous variation resembles the analysis of previous authors leveraging historical episodes to estimate the effect of population movements. Notable examples include Peters (2022) and Burchardi and Hassan (2013), both of whom examine the impact of post-WWII German migrants who resettled in West Germany. However, their estimation relies on a single event, whereas the SVAR-IV exploits changes across time. To accomplish this, I use a shift-share instrument (estimated at the shock level as stated in Borusyak, Hull and Jaravel (2022)) based on migration shares and lagged birthrates. This follows a similar identification strategy as the one proposed by previous work that estimates the effect of increasing labor supply (Shimer, 2001; Karahan, Pugsley and Şahin, 2019).

The methodology in this paper identifies exogenous shocks that influence local in-

migration flows to estimate the latter’s effect on economic conditions. In this sense, it closely resembles the Local Projection IV (LP-IV) used in [Howard \(2020\)](#), which uses a dynamic model along with an [Altonji and Card \(1991\)](#)-style instrument to determine the effect of migration on unemployment. The paper finds that unemployment falls whereas the employment-to-population ratio rises. However, it differs from the analysis I carry out in several ways. First, its econometric approach does not account for the interaction between migration, unemployment, and other relevant variables such as wages, output, and establishments, which is an element the VAR structure allows me to consider. Furthermore, it only estimates the effect on labor market outcomes, whereas this paper determines the response on a set of comprehensive economic measures. Finally, I estimate the impact on CZs which cover a broader geographical sample than MSAs, which are only representative of urban economies.

This paper also follows previous work on the implications of local dynamics for aggregate outcomes. [Davis, Fisher and Whited \(2014\)](#) study the contribution of agglomeration in cities to aggregate productivity growth through the lens of a structurally estimated spatial model. In their framework, local output per worker depends positively on output density. [Crews \(2023\)](#) further endogenizes these externalities in a heterogeneous agents model: as the aggregate stock of skill in a city increases, the rate of individual human capital accumulation rises as well¹. On the other hand, [Peters \(2022\)](#) proposes a different mechanism by relating current fixed costs of entry to past market size, creating dynamic spillovers. He embeds this in a model with variety gains, as in [Romer \(1990\)](#), and uses it to examine the scale effects on productivity in the context of postwar Germany. This paper builds on these studies by combining causal local transition responses and a theoretical framework with forward looking households to discipline the speed and strength of migration adjustment, productivity gains, and housing congestion.

The rest of the paper is structured as follows. The next section explains the SVAR-IV methodology, as well as the shock-level shift-share instrument I use to estimate the local effects of migration. Then, in [Section 3](#), I analyze the structural impulse responses to a 1% increase in inflows and discuss further supporting evidence. [Section 4](#) develops the formal framework relating the local evidence to aggregate outcomes, whereas [Section 5](#) implements the structural estimation of the model parameters. Finally [Section 6](#) implements the policy counterfactuals and [Section 7](#) concludes.

¹Previous papers in the growth literature suggested this mechanism ([Uzawa, 1965](#); [Lucas, 1988](#)), albeit at an aggregate level in a representative agent economy.

2 Econometric Model

2.1 Panel SVAR-IV Model

The first part of this paper seeks to determine the dynamic effect of population inflows on local economies. Specifically, for any commuting-zone i and year t , I estimate the impact of gross migration inflows originating in the US m_{it} on future population flows. In other words, I estimate the effect on future in- and out-migration, m_{it} and o_{it} respectively, since the arrival of new residents could drive native workers to leave as well as more immigrants to relocate in the future. Since this could be due, in part, to changes in employment opportunities, I also study the effect on the employment-to-population ratio l_{it} . This is to determine whether new workers compete for the same jobs as previous employees or if their arrival leads to additional opportunities.

Another relevant labor market outcome is the average wage w_{it} within a CZ, as an in-migration shock can increase labor supply and could therefore lead to lower wages. On the contrary, the presence of agglomeration economies can make labor productivity increase with larger market size. These two opposing effects can lead to different net impacts across different timelines. In addition to wages, I also estimate the effect on output per worker y_{it} , another measure of aggregate productivity, as well as on the net-entry rate of establishments k_{it} . Through the lens of an expanding variety model, a larger workforce leads to higher entry, which in turn increases productivity. Thus, accounting for establishments will allow me to test the predictions of such frameworks.

Finally, I determine the response of housing prices hp_{it} , a major component of households' local expenditures. As the population of an area increases, these values should increase in the short run, when the good is in fixed supply. This would in turn act as a congestion force, counteracting the agglomeration externalities and therefore leading to different effects across timelines. Moreover, recent trends in real estate prices, especially in markets experiencing a high inflow of migrants, have prompted a policy discussion aimed at increasing the housing supply. Consequently, to study these counterfactual policies we first need to estimate the effect of population increases on real estate values.

To estimate these effects, the first step is to model these variables using a panel SVAR structure, similar to the one in [Boubtane, Coulibaly and Rault \(2013\)](#), which accounts for the reciprocal relation between population flows and economic performance in a given geographical area. To this end, let $\mathbf{Y}_{it}$ denote a vector of stationary observables containing all previously defined variables in percent changes, except for the net-entry rate, which

enters in first differences.² Assume it follows a linear process:

$$\mathbf{A} \cdot \mathbf{Y}_{it} = \mathbf{c}_i + \mathbf{d}_t + \sum_{l=1}^p \boldsymbol{\alpha}_l \cdot \mathbf{Y}_{it-l} + \boldsymbol{\varepsilon}_{it} \quad (1)$$

where $\mathbf{A}$ is a non-singular matrix and $\boldsymbol{\alpha}_l \in \mathbb{R}^{7 \times 7}$ which, along with $\mathbf{A}$, describes the relation between present and past values of $\mathbf{Y}_{it}$. Furthermore, $\boldsymbol{\varepsilon}_{it}$ is a vector of structural innovations with $E[\boldsymbol{\varepsilon}_{it}] = 0$ and $E[\boldsymbol{\varepsilon}_{it}\boldsymbol{\varepsilon}_{it}'] = \mathbf{I}$. The reduced form of the model above is given by:

$$\mathbf{Y}_{it} = \boldsymbol{\mu}_i + \boldsymbol{\tau}_t + \sum_{l=1}^p \boldsymbol{\delta}_l \cdot \mathbf{Y}_{it-l} + \mathbf{B}\boldsymbol{\varepsilon}_{it} \quad (2)$$

where $\mathbf{B} = \mathbf{A}^{-1}$ and $\boldsymbol{\delta}_l = \mathbf{A}^{-1}\boldsymbol{\alpha}_l$. In the baseline results below I set $p = 3$, remove year-specific cross-sectional means and eliminate CZ fixed effects $\boldsymbol{\mu}_i$ using the forward orthogonal deviation proposed by [Arellano and Bover \(1995\)](#).

External Instrument Let $\mathbf{u}_{it} = \mathbf{B}\boldsymbol{\varepsilon}_{it}$ be the reduced-form residuals. After estimating equation (2) we need to identify $\mathbf{B}$ to construct impulse responses to structural migration shocks ε_{it}^m . Typically, when making timing assumptions, we would rely on the Cholesky decomposition of $\boldsymbol{\Sigma}$, the covariance matrix of $\mathbf{u}_{it}$, to achieve this. This is the approach [Boubtane, Coulibaly and Rault \(2013\)](#) adopt in their analysis of OECD immigration. However, since I am only interested in the response to immigration shocks, and given the ordering of variables in $\mathbf{Y}_{it}$, I only need to identify the first column $\mathbf{b}^m$. By instead using instrumental variables as detailed in [Stock and Watson \(2018\)](#), I avoid imposing any timing or sign restrictions.

To see this, rewrite the reduced-form residuals of the j^{th} VAR equation as a function of the structural migration ε_{it}^m and non-migration ε_{it}^{-m} shocks:

$$u_{it}^j = b_m^j \cdot \varepsilon_{it}^m + \mathbf{b}_{j,2:7}' \cdot \boldsymbol{\varepsilon}_{it}^{-m} \quad (3)$$

where b_m^j corresponds to the first coefficient in the j^{th} row of matrix $\mathbf{B}$. If we use a unit-effect normalization, so that $b_m^m = 1$, we can substitute ε_{it}^m with the residuals from the migration equation and obtain:

$$u_{it}^j = b_m^j \cdot u_{it}^m + e_{it}^j \quad (4)$$

where e_{it}^j is a linear combination of the non-migration structural shocks $\boldsymbol{\varepsilon}_{it}^{-m}$.

As [Stock and Watson \(2018\)](#) note, although the residuals u_{it}^j are unobserved, to estimate

²Let $\tilde{x}_{it}$ be the percentage change of variable x_{it} , then $\mathbf{Y}_{it} = [\tilde{m}_{it}, \tilde{o}_{it}, \tilde{y}_{it}, \tilde{w}_{it}, \tilde{l}_{it}, \Delta k_{it}, \widetilde{hp}_{it}]'$.

b_m^j we can use each equation in the reduced-form VAR in (2) to rewrite equation (4) as³:

$$Y_{it}^j = \mu_i^j + \tau_t^j + b_m^j \cdot \tilde{m}_{it} + \sum_{l=1}^p \Gamma_l^j \cdot Y_{it-l} + e_{it}^j \quad (5)$$

where Y_{it}^j is the j^{th} entry in $\mathbf{Y}_{it}$, and $\tilde{m}_{it}$ is the percentage change in gross migration inflows. Since the latter are correlated with non-migration shocks we generally have $E[\tilde{m}_{it}e_{it}^j] \neq 0$. Consequently, estimating the coefficients of interest requires an instrument z_{it} that satisfies the relevance and exclusion conditions, $E[\varepsilon_{it}^m z_{it}] \neq 0$ & $E[\varepsilon_{it}^{-m} z_{it}] = 0$ respectively. With this instrument, I can estimate equation (5) using two-stage least squares, recover column $\mathbf{b}^m$ and finally compute the dynamic response of all the local economic variables in $\mathbf{Y}_{it}$ to a structural population inflow shock ε_{it}^m .

2.2 Shift-Share Instruments

To identify the effect of structural market size shocks using the methodology above, I construct a set of [Altonji and Card \(1991\)](#)-style instruments z_{it}^g , based on migration shares s_{int} and demographic shocks g_{nt} :

$$z_{it}^g = \sum_n s_{int} \cdot g_{nt} \quad (6)$$

where n is the CZ from which migrants originate. Let $m_{n \rightarrow i,t}$ be the total flow of migrants coming from n to i so that $s_{int} = m_{n \rightarrow i,t-1}/N_{it-1}$. This is the “pull” factor of migration: since individuals tend to move to places that previously received migrants coming from the same area, the share s_{int} “pulls” households from n to i . On the other hand, the “push” factor will be the demographic shocks that produce an outflow from n . To this end, and following a similar strategy to [Karahan, Pugsley and Şahin \(2019\)](#) and [Shimer \(2001\)](#), I set g_{nt} equal to the 25-year lags of birth rates in n .

Note that, unlike previous migration studies that use this type of instrument, the shares are lagged by one year instead of multiple periods. Although this produces a stronger instrument it also introduces endogeneity since the predictable element of current productivity could influence past migration shares. Nevertheless, I follow [Borusyak, Hull and Jaravel \(2022\)](#) and estimate the IV regression at the *shock-level*, which in this case is at the CZ of origin level n . First, I include additional controls to equation (5). The resulting *observation-level* regression is:

$$Y_{it}^j = \mu_i^j + \tau_t^j + b_m^j \cdot \tilde{m}_{it} + \sum_{l=1}^p \Gamma_l^j \cdot Y_{it-l} + \lambda_t^j S_{it} + \sum_{r=1}^2 \rho_r^j z_{it-r} + e_{it}^j \quad (7)$$

³From (2) we know $w_{it}^j = Y_{it}^j - \sum_{l=1}^p \delta_l^j \cdot Y_{it-l} - \nu_i^j - \nu_t^j$. Use this in (4) and define $\Gamma_l^j = \delta_l^j - b_m^j \delta_l^m$ to obtain (5).

The new equation now contains two lags of z_{it} , as in [Howard \(2020\)](#), as well as an interaction between time fixed effects and the sum of migration shares S_{it} , as [Borusyak, Hull and Jaravel \(2022\)](#) suggest.

To obtain the *shock-level* specification, I regress every variable in $\mathbf{Y}_{it}$, including migration inflows $\tilde{m}_{it}$, on the controls in (7) and obtain the residual $Y_{it}^{j\perp}$. I then compute the weighted average $\bar{Y}_{it}^{j\perp} = \sum_i \frac{s_{int}}{S_{nt}} Y_{it}^{j\perp}$, where $S_{nt} = \sum_i s_{int}$. Finally, instead of estimating equation (7), I use the following specification:

$$\bar{Y}_{nt}^{j\perp} = b_j^m \cdot \bar{m}_{nt}^\perp + \bar{e}_{nt}^{j\perp}. \quad (8)$$

In addition to recovering the same coefficient of interest b_m^j , this specification has other benefits. On the one hand, it recovers the appropriate standard errors and therefore avoids the issues with classic shift-share designs shown in [Adao, Kolesár and Morales \(2019\)](#), such as the potential correlation between residuals and instruments across observations. Furthermore, this method relies on the exogeneity of the shock g_{nt} and not that of s_{int} , which is why I can use migration shares from the previous period to strengthen the instrument.

To satisfy this exclusion restriction, birth rates 25 years ago in CZs sending migrants need to be exogenous to unobservable variables affecting the current dependent variables in destinations. Specifically, following [Borusyak, Hull and Jaravel \(2022\)](#), $\bar{e}_{nt}^j$ is the average unobserved shock to variable j (for example wages) across CZs that are mostly receiving migrants from n . Lagged birth rates need to be uncorrelated with these unobservable shocks. This would be violated if, for example, individuals making fertility decisions could predict economic conditions 25 years in the future in the CZs receiving the most migrants after that time interval. Aggregate shocks, such as booms and recessions, as well as permanent productivity differences, are controlled for by time fixed effects and using differenced variables respectively.

Local Projections Estimating impulse response functions with a VAR imposes parametric restrictions on dynamics, in the sense that the joint evolution of the variables must be well approximated by a finite-lag linear system. To assess whether the estimated responses are driven by this finite-lag linear approximation VAR restriction, I also estimate impulse responses to a migration inflow shock using *observation-level* local projections:

$$\sum_{s=0}^h Y_{i,t+s}^j = \beta_{0,h}^j + \beta_h^j \tilde{m}_{it} + \mathbf{\Gamma}_h^{j'} \mathbf{X}_{it} + \mu_{i,h}^j + \tau_{t,h}^j + \lambda_{t,h}^j S_{it} + u_{h,it}^y. \quad (9)$$

where the dependent variable is the cumulative change of Y_{it}^j from $t-1$ to $t+h$. I instrument for the percent change of gross inflows $\tilde{m}_{it}$ using z_{it}^g as defined in equation (6). The controls in $\mathbf{X}_{it}$ is a vector of controls that includes two lags of the instrument as well as three lags of

all the variables in $\mathbf{Y}_{it}$. S_{it} is the sum of the shares used to build the shift-share instrument, and $\lambda_{t,h}^j$ is its year-specific slope. I include these controls to ensure comparability across both estimation methods. In Section 3.1, I show that the estimated coefficients β_h^y are comparable to the impulse responses obtained from the VAR.

2.3 Data

The IRS provides annual Statistics of Income (SOI), which include county-to-county population flows from 1990 to 2018. Specifically, for any origin i and destination j in year t , we observe the number of individual income tax returns that changed addresses from i in $t - 1$ to j in t . This dataset also includes the number of exemptions as well as the total adjusted gross income associated with those returns. Following IRS guidance, the first variable proxies for the number of households that move from one county to another whereas the second represents the number of individuals that do so. I aggregate these variables the CZ level using the the USDA ERS definitions from 2000, which map each county to one of 709 CZs. As described in Autor and Dorn (2013), these geographic areas group counties with strong commuting ties, and therefore constitute a better unit of analysis for studying the local effects of migration. Furthermore, unlike metropolitan areas that only cover certain locations, CZs encompass the entire continental US. I then define migration inflows m_{it} as the total exemption inflow from CZs within the US. I also define outflows o_{it} in a similar manner.

The BEA also provides GDP, wages, and employment at the county level, which are some of the variables in $\mathbf{Y}_{it}$ ⁴. This data is available from 2001 to 2019. Furthermore, I obtain the establishment net-entry rate for each county from the Business Dynamics Statistics (BDS) produced by the Census Bureau. The BDS also provides the different job and establishment dynamics presented in the additional results. I use the FHFA house price index, which is available at the county level, to compute an average housing index growth rate Δhp_{it} at the CZ level, weighted by the population of each county within a commuting zone. Table 1 shows the descriptive statistics for these variables. In addition to migration shares, I need birth rates at the county level to construct the instrument. I obtain this from the CDC’s National Vital Statistics microdata, which provides details on each birth during the calendar year⁵. Finally, I obtain total annual wages at the sector level for each county from the BLS Quarterly Census of Employment and Wages. This allows me to estimate the IRFs for individual industries in Section 3.2.

⁴GDP figures correspond to CAGDP2 whereas the remaining BEA variables come from the Economic Profile for counties (CAINC30)

⁵This data is publicly available for all counties until 1988. From 1989 onwards, I only have the number of births for counties with a population over 100,000 (which corresponds to 500 counties). Consequently, I impute the missing births using the average share (out of all state births) for each county for previous years.

TABLE 1 DESCRIPTIVE STATISTICS

Variable Names	Mean	St. Dev	P10	P50	P90	P95	P99
VAR variables							
Δm_{it} (%)	-1.61	32.05	-38.37	-1.06	36.01	47.97	78.45
Δo_{it} (%)	-1.62	29.20	-36.31	-0.05	32.30	43.09	68.81
Δy_{it} (%)	1.53	5.75	-3.28	1.30	6.76	10.02	19.06
Δw_{it} (%)	0.92	1.90	-0.96	0.87	2.87	3.72	6.39
Δe_{it} (%)	-0.04	2.25	-2.45	0.26	1.98	2.67	4.60
Δk_{it} (%)	-2.21	14.87	-19.26	-1.88	14.12	21.47	39.60
Δhp_{it} (%)	2.37	5.08	-3.01	2.36	7.89	10.34	16.94
Size							
Population (000s)	470.48	1,173.89	26.06	137.24	1,082.28	2,072.62	5,123.30
Employment (000s)	216.68	551.33	10.38	53.95	482.39	992.66	2,500.99
Real GDP (000s)	24,170.51	73,416.45	1,057.24	5,051.49	48,953.54	106,580.28	340,762.29
Density (Pop. per sqmi)	122.49	348.11	6.88	49.59	259.33	440.98	1144.87
Migration							
Net Inflow Rate (%)	-0.03	1.27	-0.50	-0.07	0.48	0.83	1.75
Gross Inflow Rate (%)	1.61	1.25	0.60	1.37	2.77	3.46	5.15
Gross Outflow Rate (%)	1.64	1.39	0.75	1.48	2.56	3.10	4.84
Productivity & Dynamism							
Real GDP per Worker (000s)	98.85	36.02	75.44	91.68	124.06	144.84	232.01
Real Average Wage (000s)	37.56	6.89	30.70	36.15	45.86	49.87	61.62
Employment to Population	0.42	0.07	0.33	0.42	0.52	0.54	0.60
Establishment Entry Rate	9.14	2.23	6.63	8.82	12.04	13.25	15.76

Note: Sample consists of 660 commuting zones within the continental US (excluding Alaska) and years 2002–2018.

3 The Local Effects of Migration Inflows

The first subsection estimates the SVAR-IV model described above on the full sample. The result is a set of structural impulse responses that show the evolution of local economies after a 1% increase of gross migration inflows. The remaining parts of this section delve deeper into these effects by studying the heterogeneity across different types of commuting zones as well as the impact on wages at the industry level. Throughout this section, all responses to the migration shock are measured in percent, with the exception of establishment and job-flow rates, which are in percentatge points.

3.1 Baseline Impulse Responses

The shift-share structure of the instrument allows any type of shock at the origin commuting zone level g_{nt} that is exogenous with current economic conditions. Consequently, I compare lagged birth rates with other demographic “push” factors in Table 2. The first is the total outflow from CZ n , which is a strong predictor of the specific migration flow from n to i . This helps explains why this origin-level shock produces a stronger instrument than the other two, with an F-statistic of 73.96⁶ when the shares correspond to bilateral migration normalized by total inflows. Nevertheless, under the baseline migration-to-population normalization, its F-statistic is only 0.28, and this shock is potentially highly correlated with current unobservable economic shocks affecting destinations.

Another CZ level “push” variable we can use is the total population growth rate in

⁶All the first stage results, including the F-statistic come from the *shock-level* regression developed by Borusyak, Hull and Jaravel (2022).

TABLE 2 SHOCK-LEVEL IV FIRST STAGE

	Δm_{it} (1)	Δm_{it} (2)	Δm_{it} (3)	Δm_{it} (4)	Δm_{it} (5)	Δm_{it} (6)
z_{it}^{outflow}	80.57*** (9.37)	-38.83*** (73.27)				
$z_{it}^{\text{pop. growth}}$			184.24*** (73.48)	6398.21* (3158.09)		
$z_{it}^{\text{birth rate (25y ago)}}$					0.53* (0.44)	76.78*** (16.09)
# of destination-years	8,232	8,232	8,232	8,232	8,232	8,232
# of origin-years	11242.00	11242.00	11358.00	11358.00	11358.00	11358.00
F-stat	73.96	0.28	6.29	4.10	1.47	22.79
Shares relative to total migration	✓		✓		✓	
Shares relative to total population		✓		✓		✓

Note: The dependent variable is $100\Delta \ln m_{it}$, where m_{it} denotes gross migration inflows. Before shock-level aggregation, the destination-level variables are residualized on three lags of all PVAR variables, CZ and year fixed effects, year-specific slopes on the sum of exposure shares, and two lags of the shift-share instrument. Shock-level regressions are weighted by the sum of exposure shares and report standard errors clustered by origin CZ; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Columns (1), (3), and (5) use shares relative to total migration; columns (2), (4), and (6) use migration-to-population shares. The sample covers 2002–2018.

origin n . As column (4) in Table 2 shows, the resulting instrument is highly correlated with in-migration growth, with the highest coefficient among all three shocks. The increasing population in origin CZs produces a higher outflow that gets allocated according to migration shares. Nevertheless, its F-statistic of approximately 4.10 makes this a weak instrument. Furthermore, although total population growth is more likely to satisfy the exclusion restriction than total outflow, it can still suffer from endogeneity as it is not a lagged variable.

Finally, Table 2 also reports the first stage for the instrument that uses lagged birth rates, which is the one I use to compute the baseline IRFs later in this paper. As the precisely estimated positive coefficient in column (6) shows, as z_{it}^{birth} increases so does the growth rate of migration inflows. Furthermore, this coefficient is statistically significant and possesses an F-statistic of 22.79. These results, along with the arguments provided in the previous section, make this third instrument the preferred variable for estimating the response to a structural migration shock.

TABLE 3 IMPACT COEFFICIENTS: OLS AND SHOCK-LEVEL IV

	Δo_{it}		Δy_{it}		Δw_{it}		Δl_{it}		Δk_{it}		Δhp_{it}	
	OLS (1)	IV (2)	OLS (3)	IV (4)	OLS (5)	IV (6)	OLS (7)	IV (8)	OLS (9)	IV (10)	OLS (11)	IV (12)
Δm_{it}	0.12*** (0.04)	0.10 (0.14)	0.02 (0.01)	0.14* (0.05)	0.01*** (0.00)	0.05** (0.02)	0.02*** (0.00)	0.09*** (0.02)	0.01 (0.00)	0.03** (0.02)	0.02** (0.00)	0.21*** (0.10)
# of destination-years	8232.00	8,232	8232.00	8,232	8232.00	8,232	8232.00	8,232	8232.00	8,232	8232.00	8,232
# of origin-years		11358.00		11358.00		11358.00		11358.00		11358.00		11358.00

Note: Before shock-level aggregation, the destination-level variables are residualized on three lags of all PVAR variables, CZ and year fixed effects, year-specific slopes on the sum of exposure shares, and two lags of the shift-share instrument. Shock-level regressions are weighted by the sum of exposure shares and report standard errors clustered by origin CZ; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; The sample covers 2002–2018.

Table 3 reports the shock-level two-stage least squares estimates using the birth-rate

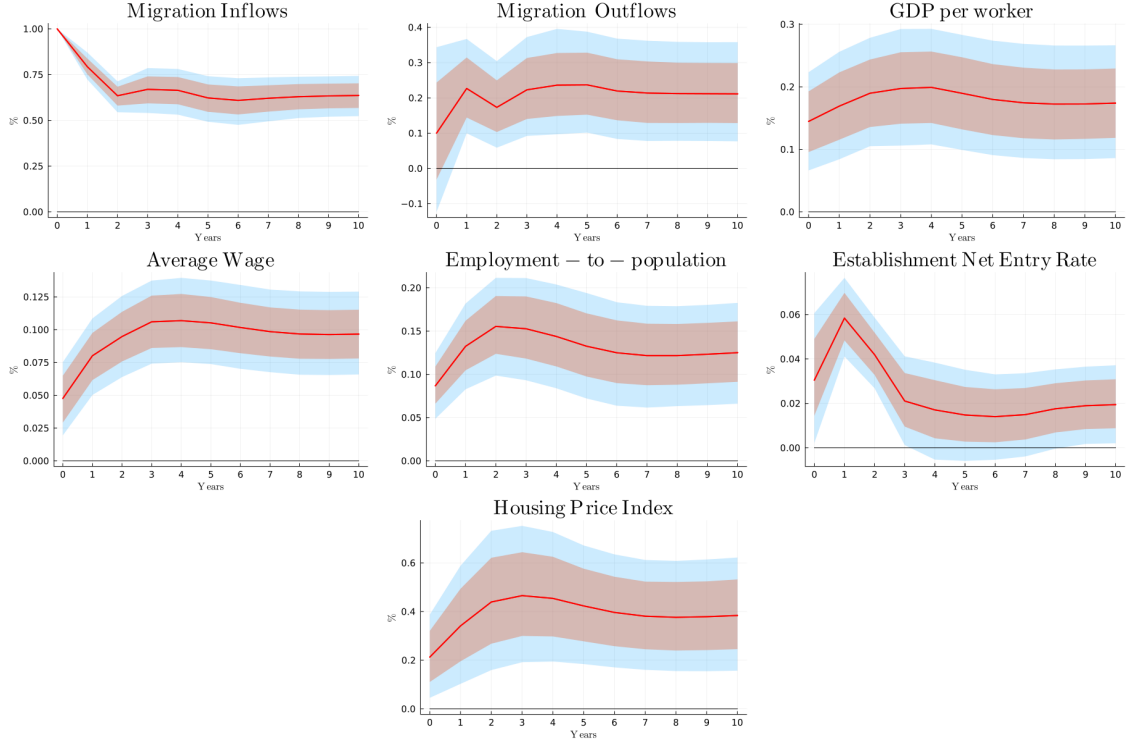
instruments. Recall that these coefficients correspond to the first column of the matrix $\mathbf{B}$ in equation (2) and therefore identify the response on impact of each outcome to a migration inflow shock ε_{it}^m , with the impact response of gross inflows normalized to one. The first result is that outflows rise on impact when inflows increase: the IV coefficient in column (2) is 0.10, but not statistically significant. In contrast, the OLS estimate in column (1) is 0.12 and statistically significant. This gap is consistent with upward bias in OLS if unobserved shocks are positively correlated with both inflows and outflows, such as destination-specific changes that simultaneously attract new migrants and induce churn among incumbent residents.

On the other hand, the contemporaneous responses of output per worker and wages are positive and statistically significant under IV, and larger than their OLS counterparts. Column (4) shows that GDP per worker rises by 0.14% on impact, while column (6) shows that average wages increase by 0.05%. These magnitudes suggest that, at least initially, the migration inflow shock acts like a positive demand shock that propagates from goods demand to labor demand. Column (8) also shows a sizable and precisely estimated rise in employment-to-population of 0.09%. Unobserved shocks that negatively affect native-dominated industries, relative to smaller ones attracting incoming migrants, could be driving the downward bias in the OLS measures. Together, these contemporaneous effects are consistent with short-run agglomeration gains, such as improved matching in thicker labor markets, operating alongside higher local demand.

Finally, column (10) indicates that net entry increases modestly on impact by 0.03 *pp*. Housing prices exhibit the largest contemporaneous response among the outcomes reported: column (12) shows they rise by 0.21% on impact. The size of the housing response is suggestive of congestion forces that counterbalance the positive effects above: when inflows raise local housing demand, prices can respond quickly in places where short-run housing supply is relatively inelastic. This immediate increase in housing costs is a natural candidate to counteract the demand-driven gains in wages and output as the adjustment unfolds. To evaluate these dynamics, I now turn to the structural impulse responses.

After increasing by 1%, gross migration inflows maintain a higher level over the horizons we observe. Figure 1 decomposes the dynamics into inflows and outflows. After the initial rise, the response of the former decreases slightly but remains persistent. It falls to 0.6–0.7% approximately over the next two years and then stabilizes near that range through year 10. After their initial increase on impact, outflows also rise following the inflow shock and remain close to 0.20% at the end of the horizon. Importantly, the increase in outflows does not indicate a deterioration of local labor-market outcomes—indeed, wages and output rise—but is consistent with turnover and sorting. As population rises, housing and other congestion-related costs increase, pushing some residents to leave in the years following the shock, even as the CZ becomes more productive.

FIGURE 1. CUMULATIVE IRFs FOR ECONOMIC VARIABLES



Note: The red line is the cumulative PVAR-IV point estimate. The darker and lighter shaded regions are the 68% and 90% percentile bands, respectively, obtained from 1,000 draws of the estimated PVAR and impact coefficients. Response of establishment net-entry rate is in percentage points. The responses of all other variables are in percent.

In the years following the inflow shock, local market size expands and economic activity responds. From the second stage results we saw output per worker increases immediately. Figure 1 shows it then continues to rise moderately, reaching its peak around year three, remaining positive through year ten at around 0.20%. Real average wages also increase after their initial response, from 0.05% on impact to roughly 0.10% after years three and four, although the magnitude is lower than the response of average productivity. Employment-to-population rises more sharply in the short run, peaking at 0.15% by year two, but then reverts slightly to 0.12% by year 10. The establishment net-entry rate experiences the fastest rise in the short run, reaching roughly 0.06 pp in year one, compared to the 0.03 pp response on impact. However, it also shows the sharpest decline, almost reaching its initial level. These patterns are consistent with short to medium-run agglomeration forces: a thicker labor market and stronger local demand can improve matching and utilization, raising measured productivity, as well as early extensive-margin expansion.

Nevertheless, housing prices also increase strongly and persistently, exerting pressure on residents' budgets. Figure 1 shows how the response of housing prices rises from 0.21% on impact to a peak of 0.46%, and remains elevated at 0.40% by year 10. Relative to output

per worker and wages, which peak at 0.20% and 0.10%, the housing-price response is larger, consistent with congestion forces that tighten local land and housing markets. Moreover, higher housing values likely proxy for a broader set of population-linked externalities, such as commuting congestion, pollution, permitting delays and higher commercial rents. These congestion costs help rationalize why outflows rise alongside inflows: some agents are priced out or choose to re-optimize away from the CZ even though wages and output per worker increase.

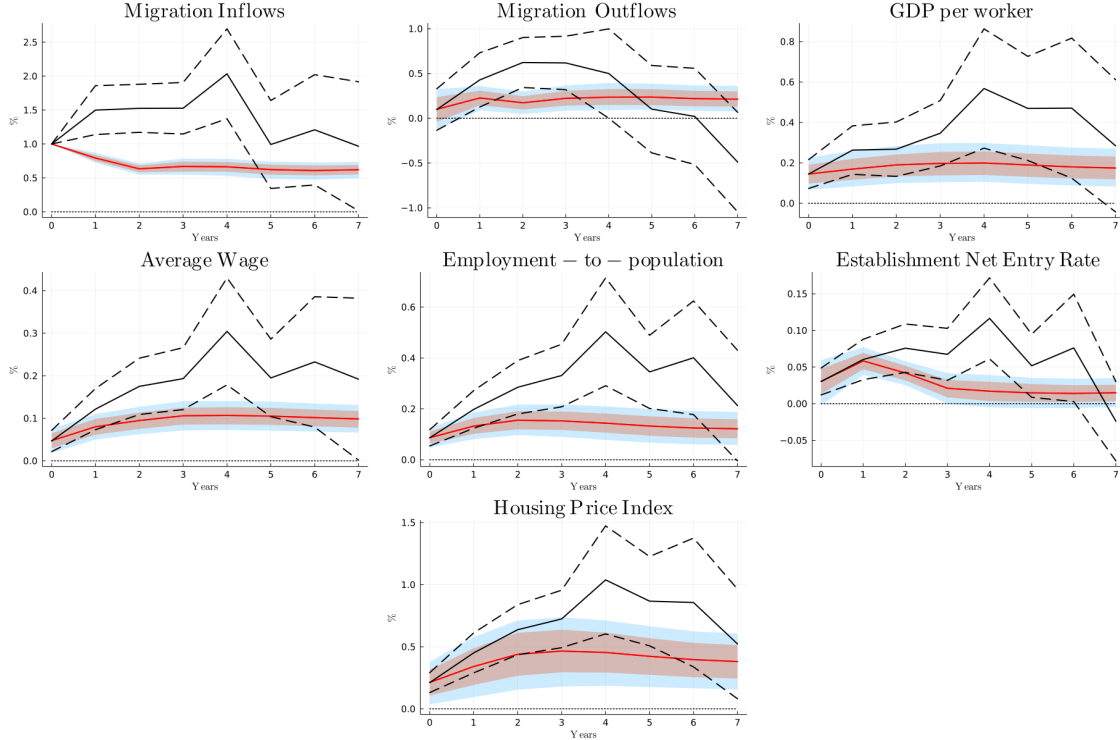
Local Projections As we saw in Section 2.1, estimating a VAR assumes we can model the joint dynamics of all variables with a finite-order lag system. Consequently, to test whether the resulting impulse responses rely on this assumption I re-estimate them using local projections. Figure 2 shows the results, along with the baseline IRFs computed with the PVAR-IV estimates. Most of the variables of interest, including average wages and output per worker, behave similarly after a migration inflow shock across both methods. The local projections generally maintain the sign of the IRFs for all variables, over the early and medium horizons. However, they are systematically larger and more volatile than the estimates obtained from the PVAR. Furthermore, following the initial increase, the LP responses start to fall approximately after year 3-4.

The local projection confidence intervals are wide and often encompass the more muted PVAR path. The latter remains the preferred specification because, under the maintained invertibility assumptions, it estimates the joint dynamics of the system; by contrast, the former serves as a robustness check that avoids imposing the recursive propagation inherent in a VAR. Both approaches yield the same impact effects and main qualitative conclusions: following a positive inflow shock, CZs experience an expansion in economic activity that leads to higher output, wages, employment, and housing prices. Quantitatively, the LP-IV effects are larger and less precisely estimated over the medium and longer term. However, this pattern also reflects the response of migration; when each outcome is normalized by the corresponding inflow response, the dynamic effects across the two estimators become remarkably similar. Ultimately, this comparison suggests that the primary discrepancy between the two methods lies in the persistence of the inflow shock rather than the direction of its local economic effects.

3.2 Additional Dynamics and Heterogeneous Effects

As inflows rise, a commuting zone sees persistent increases in output per worker, wages, employment and housing prices, while net-entry of establishments exhibits a temporary increase. To determine whether the expansion occurs through new entrants or incumbents, I next examine the response of additional establishment and job-flow measures, before looking at sectoral and cross-CZ heterogeneity.

FIGURE 2. PVAR-IV AND LOCAL-PROJECTION IV CUMULATIVE RESPONSES



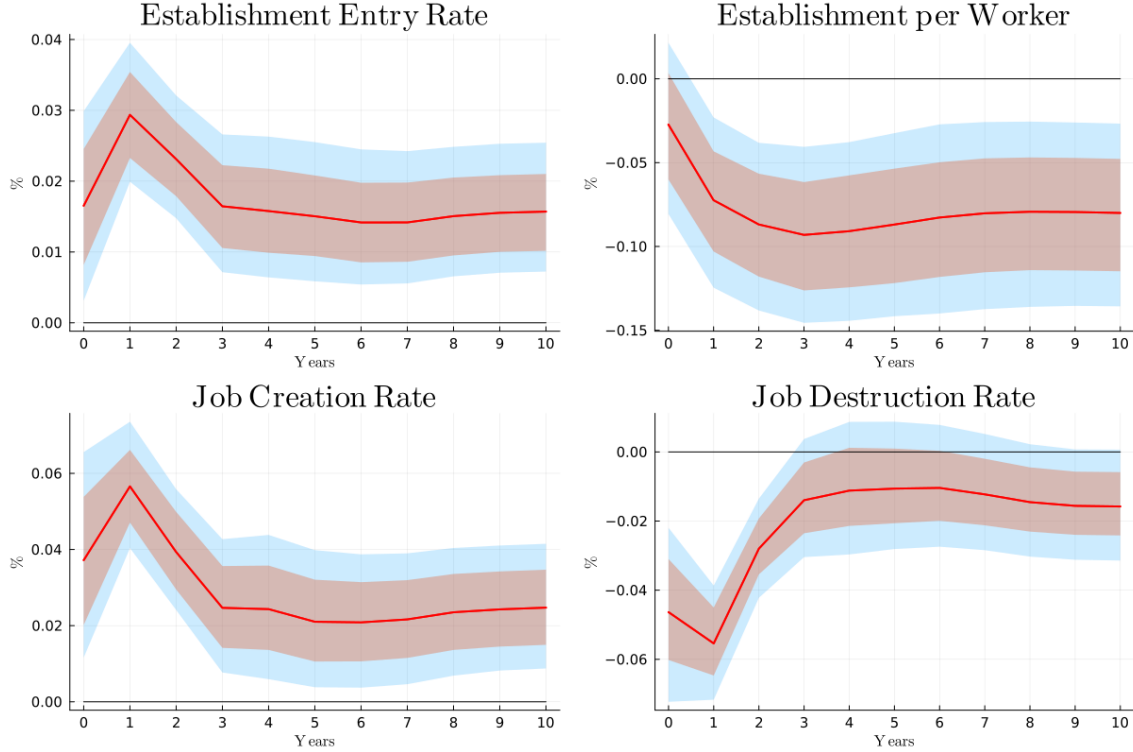
Note: The red line is the cumulative PVAR-IV response; the darker and lighter shaded regions are its 68% and 90% percentile bands based on 1,000 parameter draws. The solid black line is the LP-IV estimate, and the dashed black lines are its 90% interval, constructed as the estimate plus or minus 1.65 CZ-clustered standard errors. The comparison is shown through year 7, the longest LP horizon.

Establishment Dynamics and Job Creation In the baseline results, the establishment net-entry rate increases after a positive inflow shock, peaking after one year before falling back approximately to pre-shock levels. I now augment the VAR by adding different measures of dynamism one at a time.⁷, starting with the gross entry rate. The rise we see in Figure 3 shows that most of the increase in the net-rate comes from a rise in number of establishments entering as the gross establishment entry rate peaks near 0.03 percentage points in year one and remains about 0.015 percentage points above baseline in year 10. At the same time, establishments per worker decline by roughly 0.09% by year three and remain below baseline thereafter. Thus, even though the establishment counts increase, to accommodate the growing labor force the average establishment has to grow in size.

Finally, the job-flow responses reinforce this same narrative. Job destruction initially drops sharply—by almost 0.06 percentage points in year one—before quickly recovering to level off between 0.01 and 0.02 percentage points below its pre-shock level. Gross job

⁷For the gross-entry and job-flow panels in Figure 3, I add the displayed variable to the baseline seven-variable system and re-estimate the PVAR-IV. For the establishments-per-worker panel, I replace the establishment net-entry rate with establishments per worker. Each panel therefore comes from a separate specification.

FIGURE 3. CUMULATIVE IRFS FOR ESTABLISHMENTS AND JOBS

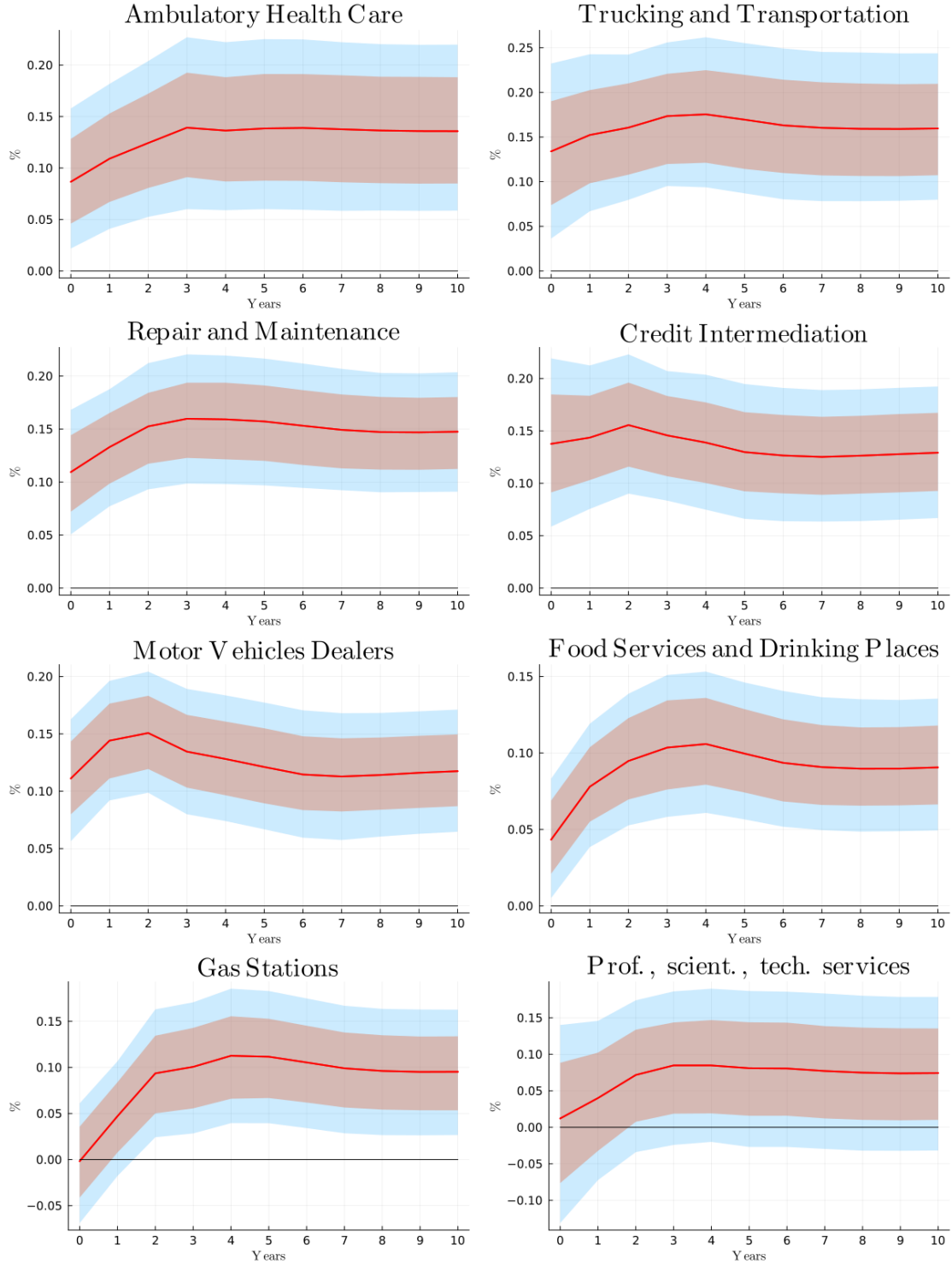


Note: The red line is the cumulative PVAR-IV point estimate. The darker and lighter shaded regions are the 68% and 90% percentile bands, respectively, obtained from 1,000 draws of the estimated PVAR and impact coefficients. Response of establishment gross-entry and job-flows are in percentage points. The response of establishments per worker is in percentage.

creation follows the opposite path: it rises by about 0.06 percentage points in year one and remains positive thereafter. These impulse responses point to an adjustment across both margins over time: an initial expansion driven by new entrants and net job creation, followed by a period of lower employment turnover and incumbent growth. This sequence highlights the importance of evaluating the joint dynamics of these variables following a migration inflow shock.

Wage Effect by Sector If migrant income is positively correlated with the instrument, then the identifying variation might be driven by relatively high-income migrants, part of the baseline wage response in Figure 1 could reflect composition. To assess whether wage effects are broader than composition, I examine wage responses within industries that employ relatively homogeneous labor in terms of skill. If wages move within these sectors, that suggests in-migration affects local pay more generally rather than only changing who lives in the commuting zone. Although this does not discard within-industry composition changes, it provides a useful check on whether the aggregate result is driven only by reallocation across sectors. To do so, I select a set of 3-digit NAICS industries and re-estimate the

FIGURE 4. CUMULATIVE WAGE RESPONSES ACROSS SELECTED NAICS INDUSTRIES



Note: The red line is the cumulative PVAR-IV point estimate. The darker and lighter shaded regions are the 68% and 90% percentile bands, respectively, obtained from 1,000 draws of the estimated PVAR and impact coefficients. All responses are in percentage.

VAR multiple times, each time adding the average wage for one industry as an additional outcome, analogous to the exercise with job dynamics.⁸ Figure 4 reports the resulting impulse responses.

The results suggest the baseline wage increase is not purely a composition effect, driven uniquely by shifts toward higher-paying industries. For reference, the aggregate wage response in Figure 1 rises modestly in the short run and peaks around 0.10% over the first few years in the baseline specification. Figure 4 shows wages also rise within several lower-skill, locally oriented industries. Trucking and transportation has the largest response, at about 0.18%, followed by repair and maintenance, credit intermediation, and motor-vehicle dealers at roughly 0.15–0.16%. These responses are even higher than the baseline result and are consistent with an expansion in local market size that raises demand for non-tradable services and strengthens matching in thick labor markets. This in turn generates short and long-run agglomeration gains that translate into higher pay for some workers.

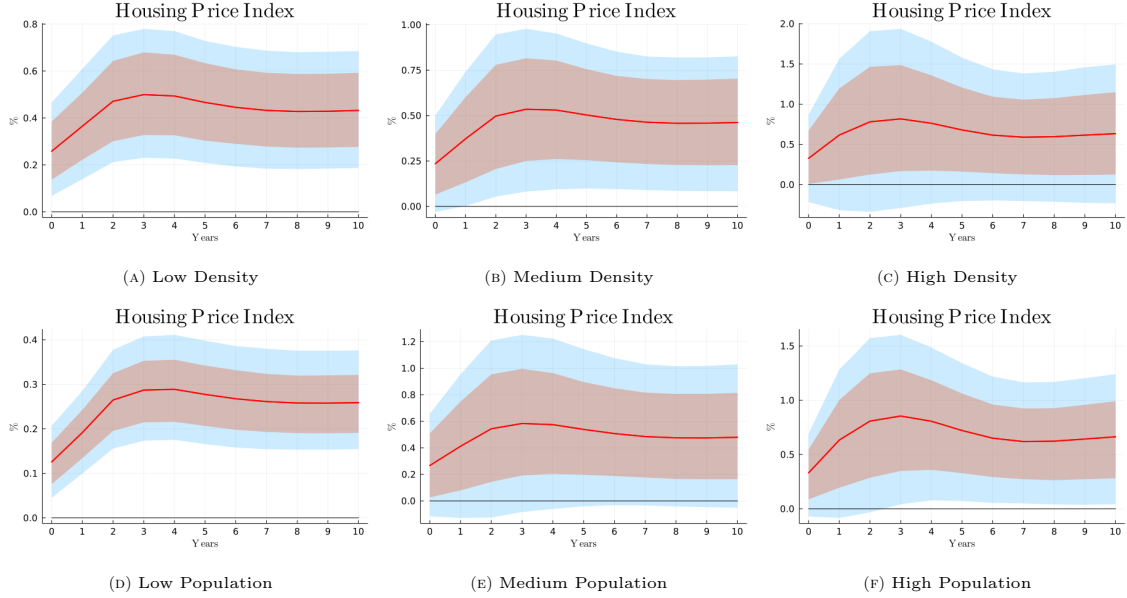
Heterogeneity by CZ Density and Population I now study heterogeneity across commuting zones (CZs), focusing on output per worker and housing prices, as proxies for agglomeration and congestion externalities respectively. The reason different densities could produce heterogeneous responses is the differential relevance of congestion and agglomeration forces. Workers in NYC interact with each other more frequently than those in Phoenix, where land is less densely populated, which facilitates the human capital externalities that Lucas (1988) described. Nevertheless, the former also experiences higher congestion. For each CZ, I compute average population and population density over the sample and classify locations into low, middle, and high groups using the 33rd and 66th percentiles. I then re-estimate the SVAR-IV model separately for each group.

In Figure 5, I report the resulting impulse responses for housing prices by CZ group. Housing prices serve as a natural proxy for congestion costs: when inflows raise local housing demand, price pressure should be stronger where short-run housing supply is inelastic and weaker where supply can adjust more easily. Because supply tightness is closely linked to density and market size, the housing-price response is expected to vary systematically across these groups. Indeed, although prices rise across all CZ types, the magnitude of the response differs by density and population. Whereas low-density CZs exhibit a peak increase of approximately 0.5%, high-density CZs attain a level that is almost 1% higher than its pre-shock baseline. A similar difference emerges between low- and high-population CZs, where prices in the former rise by only 0.3%, compared with the latter, where the increase also approaches 1%.

Whereas housing-price appreciation is strongest in high-density areas, the point esti-

⁸The Quarterly Census of Employment and Wages (QCEW) provides average wages at the county-industry level. To aggregate them at the CZ level, I use the same delineation detailed in Section 2. Each industry is therefore associated with its own VAR specification.

FIGURE 5. HOUSING PRICE RESPONSE BY CZ DENSITY AND POPULATION

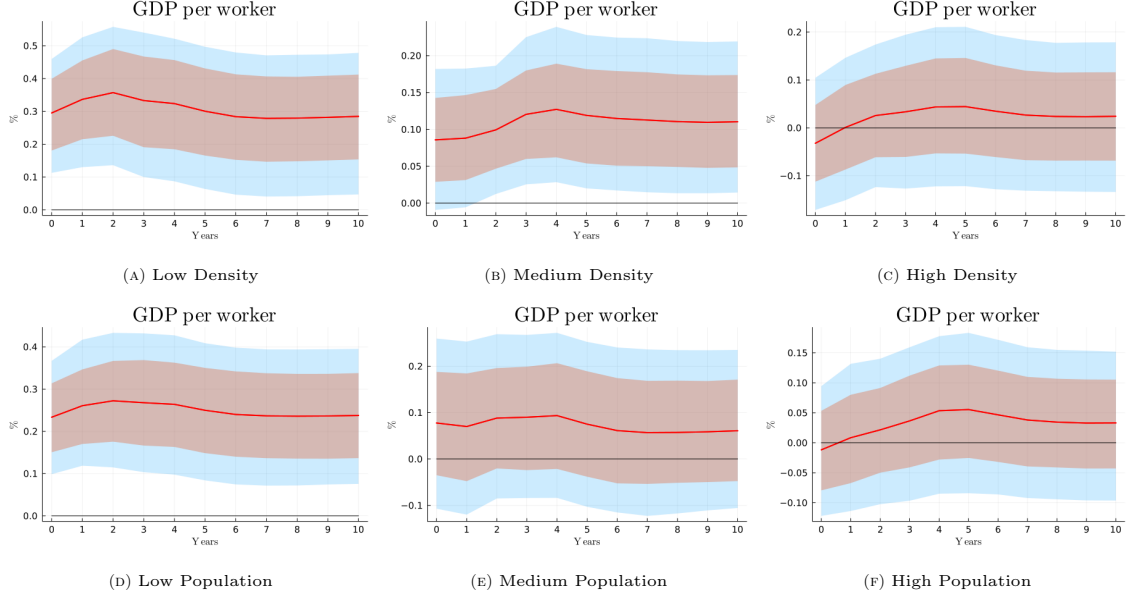


Note: The red line is the cumulative PVAR-IV point estimate. The darker and lighter shaded regions are the 68% and 90% percentile bands, respectively, obtained from 1,000 draws of the estimated PVAR and impact coefficients. All responses are in percentage.

mates for output per worker display the opposite ranking. Figure 6 shows that real GDP per worker increases by approximately 0.35% two years after the migration shock in low-density areas, whereas the response in medium-density CZs reaches a peak close to 0.12% after three years. Notably, high-density locations do not experience a statistically significant response in average productivity following higher population inflows. A similar pattern emerges across CZs of different sizes: low-population CZs experience an immediate increase that remains between 0.20% and 0.30% in the years following the shock, whereas for both medium- and high-population CZs, output per worker remains close to its baseline level.

Taken together with the heterogeneous housing-price responses, these estimates suggest that smaller and less dense markets may have greater scope to realize productivity gains from a marginal expansion in market size. Such gains could operate through market-size effects—including stronger local demand—as well as agglomeration mechanisms, such as improved matching. In larger and denser CZs, by contrast, output-per-worker gains are muted while housing-price pressures are stronger. Although these results do not separately identify the underlying mechanisms, they show that the net balance between agglomeration and congestion varies systematically with local market size and density.

FIGURE 6. OUTPUT PER WORKER RESPONSE BY CZ DENSITY AND POPULATION



Note: The red line is the cumulative PVAR-IV point estimate. The darker and lighter shaded regions are the 68% and 90% percentile bands, respectively, obtained from 1,000 draws of the estimated PVAR and impact coefficients. All responses are in percentage.

4 Theoretical Model

The impulse responses demonstrate how higher migration inflows produce a positive and sustained expansion in local economic outcomes, including wages, output, and housing prices. Although the direction of these effects is uniform across commuting zones of varying population sizes and densities, their magnitudes display clear heterogeneity. While wage and productivity gains capture the strength of agglomeration externalities and labor demand, rising housing prices proxy for congestion forces. To quantify the strength of these competing mechanisms, this section develops a continuous-time spatial model featuring static gains from a larger mass of firms alongside dynamic spillovers between current market size and future entry costs. The model also incorporates a fixed land endowment for housing and population-dependent amenities, which render locations less attractive as their size increases. The structural parameters of this model are subsequently disciplined using the impulse responses estimated above.

4.1 Environment

Households The economy consists of regions $g \in G$ in continuous time $t \in \mathbb{R}_+$ and a mass $N(t)$ of workers. Workers discount the future at rate ρ and exit the labor force at the common rate δ . New workers enter region g at the flow $E_g(t) \equiv b_g(t)N_g(t)$ where $b_g(t)$ is the region-specific “birth rate”. Upon entry, workers draw a permanent raw skill $z \in \Omega_z$

from the distribution $F(z)$ with density $f(z)$. Region g converts raw skill into effective labor through the location-specific return-to-skill shifter

$$a_g(z) = z \cdot e^{\{\chi_g(\log z - \mu_z)\}}. \quad (10)$$

so a worker with raw skill z supplies $a_g(z)$ units of labor in region g ⁹. This implies the return to skill varies across locations, and captures how the productivity of workers with the same raw skill can differ across places due to local technologies, tasks, and industry composition. The modeling choice here is a reduced form analogue of the mechanism in [Bryan and Morten \(2019\)](#), where workers draw idiosyncratic location-specific productivity.

Households derive utility from consumption $C_g(t)$ and housing $H_g(t)$, as well as an amenity $D_g(t)$ so that:

$$U_g(C_g(t), H_g(t)) = D_g(t) \cdot C_g(t)^\alpha \cdot H_g(t)^{1-\alpha}$$

where $D_g(t) = D_g \cdot N_g(t)^{-\phi}$ and $\phi > 0$. Note that amenities are decreasing with local population. This is to account for congestion externalities other than housing such traffic or the public goods. The consumption good is a CES aggregate from differentiated regional varieties $C_g = (\sum_r c_{rg}^{\frac{\sigma-1}{\sigma}})^{\frac{\sigma}{\sigma-1}}$. To buy variety r in g , households need to pay $p_{rg}(t)$, whereas $P_g^h(t)$ is the price for each unit of housing they consume.

Each period thereafter they receive a migration opportunity, whose arrival follows a Poisson process with rate λ ¹⁰. This parameter captures the fixed costs associated with moving to new markets. This is one of the two mobility frictions the model includes. After receiving the migration opportunity, workers draw a preference shock for each location $\xi_g(t)$, which are distributed i.i.d. Frechet with shape parameter ν . The second friction is a flow-utility cost κ_{gd} that the household pays for moving from g to d , where $\kappa_{gd} < 1$ for $g \neq d$ and $\kappa_{gg} = 1$. Thus workers consider the value of living in each location, given these mobility costs and preference shocks that are both multiplicative and permanent as in [Crews \(2023\)](#).

The mobility decisions of workers, along with their entry and exit from the economy, will determine the spatial distribution of z at time t denoted by $\psi_g(z, t)$. Regional population and effective labor are therefore

$$N_g(t) = N(t) \int_{\Omega_z} \psi_g(z, t) dz, \quad L_g(t) = N(t) \int_{\Omega_z} a_g(z) \psi_g(z, t) dz. \quad (11)$$

Workers with raw skill z supplies effective labor $a_g(z)$ inelastically in the region g they choose to live in, and thus earn total labor income $a_g(z)W_g(t)$.

⁹I center log skill by μ_z , where $\mu_z = \int_{\Omega_z} \log z f(z) dz$. This makes χ_g a return-to-skill slope rather than a pure level shifter.

¹⁰I assume that newborns do not receive this migration opportunity and are not subject to the exit probability.

Production Perfectly competitive local producers supply housing services $H_g(t)$ by combining land $T_g(t)$ and labor $L_g^h(t)$ as inputs using the following function:

$$H_g(t) = \Gamma_h T_g(t)^\theta (L_g^h(t))^{1-\theta}.$$

where $\Gamma_h = \theta^{-\theta}(1-\theta)^{-(1-\theta)}$. I assume land is a fixed factor owned by immobile landlords who collect rents $R_g(t)$.

Following [Romer \(1990\)](#), production of regional varieties is subject to variety gains. Each period, under perfect competition, a local firm sources $y_{gd}(j, t)$ at price $q_g(j, t)$ from firm j to produce y_{gd} , the amount of variety $g \in G$ it ships to destination d . These inputs are aggregated following:

$$y_{gd}(t) = \frac{1}{\tau_{gd}} \left(\int_0^{M_g(t)} y_{gd}(j, t)^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}$$

where $M_g(t)$ is the mass of intermediate input producers in g at time t , and τ_{gd} represents the usual iceberg trade cost. Local intermediate producers on the other hand face monopolistic competition as well as labor requirements given by

$$l_g(j, t) = \frac{y_g(j, t)}{A_g} + F_g(t)$$

where A_g is the fixed productivity in region g and $F_g(t)$ is the fixed cost in terms of labor units that firms have to pay. Similar to the discrete time setting in [Peters \(2022\)](#), the growth rate of this cost depends on the mass of firms as well as the *inter-temporal knowledge elasticity* η :

$$\gamma_{F_g}(t) \equiv \frac{\dot{F}_g(t)}{F_g(t)} = \log \left(\frac{1}{F_g(t) \cdot M_g(t)^\eta} \right). \quad (12)$$

The parameter η controls the dynamic spillover from the current mass of firms to future entry costs. A larger $M_g(t)$ lowers the growth rate of $F_g(t)$ and can therefore generate persistent agglomeration effects.

4.2 Static Equilibrium

At any point in time t , the aggregate state variable is total population $N(t)$, whereas the regional predetermined objects are the regional skill-location densities $\{\psi_g(z, t)\}_{g \in G}$, and the fixed entry costs $\{F_g(t)\}_{g \in G}$. These objects then determine local population and labor through (11), while the mass of firms adjusts through the free entry equation (15). We can therefore define a static equilibrium as follows:

Definition 1. For each $g \in G$, given $F_g(t)$ and $\psi_g(z, t)$, a **static equilibrium** in period $t \in \mathbb{R}_+$ is a set of prices and allocations for skill $\{W_g(t), L_g(t)\}$, consumption imported

from all regions $\{c_{rg}(t), p_{rg}(t)\}_{r \in G}$, intermediate inputs $\{q_g(j, t), y_{gd}(j, t)\}_{j \in M_g(t), d \in G}$, housing $\{H_g(t), P_g^h(t)\}$ as well as land $\{T_g(t), R_g(t)\}$ such that given prices i) households and firms behave optimally, ii) allocations clear their respective markets.

Given their place of residence, households decide how much to consume of each region's variety with a local price index given by $P_g(t) = (\sum_r p_{rg}(t)^{1-\sigma})^{1/(1-\sigma)}$. Worker z will spend a constant share of her income $v_g(z, t) = a_g(z) \cdot W_g(t)$ on consumption goods and housing services. Since land is fixed the supply of housing services under perfect competition will be:

$$H_g(t) = \left(\frac{P_g^h(t)}{W_g(t)} \right)^{\frac{1-\theta}{\theta}} \cdot \frac{T_g}{\theta} \quad (13)$$

On the other hand, local producers decide how much of their output they allocate to each region (including their own) as well as the demand for local varieties, which will be

$$y_g(j, t) = \left(\frac{Q_g(t)}{q_g(j, t)} \right)^\varepsilon Y_g(t),$$

where $Y_g(t)$ is the total output of region g and $Q_g(t)$ is the price index these firms face. Intermediate producers take this demand and choose prices and labor so as to charge a markup over marginal cost $q_g(j, t) = \frac{\varepsilon}{\varepsilon-1} \cdot \frac{W_g(t)}{A_g}$, which implies $Q_g(t) = M_g(t)^{\frac{1}{1-\varepsilon}} \left(\frac{\varepsilon}{\varepsilon-1} \cdot \frac{W_g(t)}{A_g} \right)$. Furthermore, free entry determines total labor demand for production in region g :

$$L_g^y(t) = \int_0^{M_g(t)} l_g(j, t) dj = M_g(t) F_g(t) \varepsilon \quad (14)$$

Market clearing for labor states demand from the goods and housing sector has to equal total supply $L_g(t)$. This, along with zero profits and housing market clearing, sets the total efficiency units in goods production as a constant share of regional labor: $L_g^y(t) = [\alpha + \theta(1 - \alpha)] L_g(t)$. Combining this expression with free entry yields the mass of active intermediate producers:

$$M_g(t) = \frac{[\alpha + \theta(1 - \alpha)] L_g(t)}{\varepsilon F_g(t)}, \quad (15)$$

which depends on the dynamics of the fixed cost $F_g(t)$.

Total income in region g is $W_g(t) L_g(t)$ and balanced trade implies:

$$W_g(t) L_g(t) = \sum_d \pi_{gd}(t) W_d(t) L_d(t) \quad (16)$$

where $\pi_{gd}(t)$ is the share of consumption expenditures in d spent in goods from g . This in

turn is given by:

$$\pi_{gd}(t) = \frac{\left(\tau_{gd} \cdot Q_g(t)\right)^{1-\sigma}}{\sum_{i \in G} \left(\tau_{id} \cdot Q_i(t)\right)^{1-\sigma}} \quad (17)$$

Using equilibrium input prices I obtain the price index $P_g(t) = Q_g(t) \cdot \pi_{gg}(t)^{\frac{1}{\sigma-1}}$ as well as the price of housing units in region g given by:

$$P_g^h(t) = \left(\frac{(1-\alpha) \cdot L_g(t) \cdot \theta}{T_g}\right)^\theta \cdot W_g(t) \quad (18)$$

Combining them with the demand for consumption goods and housing, I derive the flow utility a worker with productivity level z receives when living in region g :

$$U_g(z, t) = D_g N_g(t)^{-\phi} a_g(z) \frac{\Gamma_g}{\Gamma_U} (\pi_{gg}(t))^{\frac{\alpha}{1-\sigma}} (M_g(t))^{\frac{\alpha}{\varepsilon-1}} (L_g(t))^{-\theta(1-\alpha)}, \quad (19)$$

where $\Gamma_U = (\frac{\varepsilon}{\varepsilon-1})^\alpha ((1-\alpha)\theta)^{\theta(1-\alpha)}$ and $\Gamma_g = A_g^\alpha T_g^{\theta(1-\alpha)}$. Note the real wage is a function of the mass of firms and total efficiency units in the regional market, as well as those in other localities through the own trade share $\pi_{gg}(t)$.

4.3 Dynamic Equilibrium and Balanced Growth Path

Now that we defined a static equilibrium for every t we can study the dynamics of the model, which involves spatial labor reallocation through migration as well as the evolution of the mass of firms within each region.

The following Hamilton-Jacobi-Bellman (HJB) equation summarizes the household's recursive problem (I derive this equation in Appendix A):

$$(\delta + \rho)V_g(z, t) = U_g(z, t) + \dot{V}_g + \lambda \sum_i m_{gi}(z, t) \left[\delta_{gi}(z, t) V_i(z, t) - V_g(z, t) \right] \quad (20)$$

where $\delta_{gi}(z, t) = \frac{1}{G} \cdot \kappa_{gi} \cdot m_{gi}(z, t)^{-(\frac{1}{\nu}+1)}$ and the migration shares are given by:

$$m_{gd}(z, t) = \frac{[\kappa_{gd} \cdot V_d(z, t)]^\nu}{\sum_i [\kappa_{gi} \cdot V_i(z, t)]^\nu} \quad (21)$$

This is the probability that a worker with productivity z moves from region g to d , conditional on having a migration opportunity mediated by λ .

Recall that the distribution of individual productivities $\psi_g(z, t)$ will determine regional labor and population. Thus the evolution of these variables will depend on the Kolmogorov

forward equation below (see Appendix A for a derivation):

$$\begin{aligned}
\dot{\psi}_g(z, t) &= b_g(t) \cdot \frac{N_g(t)}{N(t)} \cdot f(z) - \delta \psi_g(z, t) \\
&- \lambda \left[(1 - m_{gg}(z, t)) \psi_g(z, t) - \sum_{i \neq g} m_{ig}(z, t) \psi_i(z, t) \right] \\
&- \psi_g(z, t) \frac{\dot{N}(t)}{N(t)}
\end{aligned} \tag{22}$$

Finally, equation (15) and (12) establish a relation between the fixed cost, labor and the mass of firms. Namely, given a fixed entry cost $F_g(t)$, which evolves according to equation (12), the mass of firms is determined by the free-entry condition (15). Holding regional labor fixed, a lower fixed cost supports a larger mass of firms rather than a lower aggregate demand for production labor. This is the model's dynamic agglomeration channel: a larger current market lowers future entry costs, which allows more varieties to operate subsequently. I can now define a dynamic equilibrium.

Definition 2. A *dynamic equilibrium* is a collection of value functions $V_g(z, t)$, migration shares $m_{gd}(z, t)$, skill-location densities $\psi_g(z, t)$, fixed costs $F_g(t)$, firm masses $M_g(t)$, regional populations $N_g(t)$, effective labor supplies $L_g(t)$, and wages $W_g(t)$ such that, for every $g, d \in G$, $z \in \Omega_z$, and $t \in \mathbb{R}_+$:

1. fixed costs evolve according to (12), and firm masses satisfy the free-entry condition (15);
2. V_g and m_{gd} solve (20) and (21), taking aggregate paths as given;
3. ψ_g evolves according to (22);
4. population and effective labor satisfy (11), with $N(t) = \sum_g N_g(t)$ and $\dot{N}(t) = \sum_g E_g(t) - \delta N(t)$;
5. the resulting prices and allocations constitute a static equilibrium at every date.

Balanced Growth Path A balanced growth path (BGP) is a special type of dynamic equilibrium where all variables grow at a constant rate, and detrended regional distributions and allocations are stationary. To define such a solution, I detrend each variable $x(t)$ by rewriting it relative to its long-run growth rate along the BGP $x(t) = \tilde{x}(t) \cdot e^{\gamma_x t}$. For all derivations see Appendix A. The result is a set of detrended equilibrium conditions.

Free entry implies $\gamma_M = \gamma_L - \gamma_F$. Combining this condition with the fixed-cost process in equation (12) gives $\gamma_M = \frac{1}{1-\eta} \gamma_L$, thereby connecting the growth rate of $M_g(t)$ and that

of $L_g(t)$ along the BGP. I also detrend the HJB in equation (20) to obtain:

$$\begin{aligned}
(\delta + \rho - \gamma_v)\tilde{V}_g(z, t) &= \tilde{U}_g(z, t) + \partial_t \tilde{V}_g(z, t) \\
&+ \lambda \sum_i m_{gi}(z, t) \left[\delta_{gi}(z, t) \tilde{V}_i(z, t) - \tilde{V}_g(z, t) \right],
\end{aligned} \tag{23}$$

where the effective discount rate is now $(\delta + \rho - \gamma_v)$ with $\gamma_v = \frac{\alpha}{\varepsilon-1}\gamma_M - [\theta(1-\alpha) + \phi]\gamma_L$.

The detrended Kolmogorov forward equation (22) is now:

$$\begin{aligned}
\partial_t \tilde{\psi}_g(z, t) &= b_g(t) \frac{\tilde{N}_g(t)}{\tilde{N}(t)} f(z) - \delta \tilde{\psi}_g(z, t) \\
&- \lambda \left[(1 - m_{gg}(z, t)) \tilde{\psi}_g(z, t) - \sum_{i \neq g} m_{ig}(z, t) \tilde{\psi}_i(z, t) \right] \\
&- \left(\tilde{\gamma}_N(t) + \gamma_N \right) \tilde{\psi}_g(z, t).
\end{aligned} \tag{24}$$

By detrending this equation I find $\gamma_\psi = 0$, which implies from the definition of $L_g(t)$, that labor grows at the same rate as population along the BGP so that $\gamma_L = \gamma_N$. The detrended versions of these variables are defined by

$$\tilde{N}_g(t) = \tilde{N}(t) \int_{z \in \Omega_z} \tilde{\psi}_g(z, t) dz \quad \& \quad \tilde{L}_g(t) = \tilde{N}(t) \int_{z \in \Omega_z} a_g(z) \tilde{\psi}_g(z, t) dz.$$

Using population dynamics, I compute the growth rate of $\tilde{N}_g(t)$ as a function of regional birth rates and δ :

$$\frac{\dot{\tilde{N}}(t)}{\tilde{N}(t)} = \frac{\sum_g b_g(t) \cdot \tilde{N}_g(t)}{\tilde{N}(t)} - \delta - \gamma_N \tag{25}$$

Note that once the overall economy reaches the BGP, all detrended variables $\tilde{x}(t)$ are constant so that $\tilde{x}(t) = \bar{x} \forall t$. I now formally define this type of equilibrium below.

Definition 3. A *balanced growth path* is a dynamic equilibrium with stationary detrended distributions and constant growth rates such that

$$\begin{aligned}
V_g(z, t) &= e^{\gamma_v t} \bar{V}_g(z), \quad m_{gd}(z, t) = \bar{m}_{gd}(z), \quad \psi_g(z, t) = \bar{\psi}_g(z), \\
F_g(t) &= e^{\gamma_F t} \bar{F}_g, \quad M_g(t) = e^{\gamma_M t} \bar{M}_g, \quad L_g(t) = e^{\gamma_N t} \bar{L}_g, \\
N_g(t) &= e^{\gamma_N t} \bar{N}_g, \quad N(t) = e^{\gamma_N t} \bar{N}.
\end{aligned}$$

The growth rates satisfy $\gamma_M = \frac{1}{1-\eta}\gamma_N$ and $\gamma_v = \frac{\alpha}{\varepsilon-1}\gamma_M - [\theta(1-\alpha) + \phi]\gamma_N$, while the stationary objects solve the BGP HJB, Kolmogorov, free-entry, trade-balance, and market-clearing conditions.

Along the BGP, the detrended HJB (23) becomes:

$$(\delta + \rho - \gamma_v)\bar{V}_g(z) = \bar{U}_g(z) + \lambda \sum_i \bar{m}_{gi}(z) \left[\bar{\delta}_{gi}(z)\bar{V}_i(z) - \bar{V}_g(z) \right] \quad (26)$$

whereas the Kolmogorov equation (24) is now

$$0 = \bar{b}_g \frac{\bar{N}_g}{\bar{N}} f(z) - (\delta + \gamma_N)\bar{\psi}_g(z) - \lambda(1 - \bar{m}_{gg}(z))\bar{\psi}_g(z) + \lambda \sum_{i \neq g} \bar{m}_{ig}(z)\bar{\psi}_i(z) \quad (27)$$

Finally, to find the population growth rate γ_N , I use the fact that $\dot{\bar{N}} = 0$ along the BGP in equation 25. This implies:

$$\gamma_N = \frac{\sum_g \bar{b}_g \bar{N}_g}{\bar{N}} - \delta. \quad (28)$$

Note that I could also derive this by integrating the BGP Kolmogorov equation (27) over all z and summing over g , as well as using the fact that along the BGP there is no net migration.

4.4 Model Solution

To compute the solution to this model, I adapt the algorithms proposed by Achdou et al. (2022). A similar approach was implemented in a spatial setting by Crews (2023) to compute the steady state for a centralized economy. However, because estimating the structural parameters via impulse response matching requires simulating dynamic outcomes, I implement a fixed-point procedure that solves both the economy's transition path and its BGP. To compute the the latter, I discretize the raw-skill space and solve for the BGP as a fixed point in regional effective labor, relative wages, and population. For the transition path, I take the initial and terminal BGPs, alongside a trajectory of regional entry rates, as given. These entry rates correspond to the exogenous birth rates used to construct the simulated instrument. I then solve for the fixed point across the entire path of endogenous variables. To simplify notation, I collect all global parameters into Θ and regional fundamentals into Θ_G . Appendix B provides the details of the numerical discretization and convergence criteria.

5 Structural Estimation

With the framework in the previous section I can study the effects of local migration and relocation shocks on aggregate outcomes such as output and productivity. To carry this out, I first estimate the structural parameters so that the model is consistent with the impulse responses in Section 3.

5.1 Estimation Strategy

The model parametrization can be separated into two groups. The first includes parameters that are calibrated before solving either the BGP or the transition of the model. This includes common parameters such as ρ , α , $\bar{N}$, γ_L , as well as those controlling the skill distribution F . The elasticity of trade costs to distance, ζ_τ , which I describe in more detail below, is also in this group. I also calibrate some regional parameters before solving the model, namely regional usable land T_g , return-to-skill shifter χ_g and the path of lagged birth rates $b_g(t)$.

The second group can be divided further into two categories. On the one hand, for every candidate structural vector, I calibrate the regional productivities A_g , amenities D_g , bilateral migration costs κ_{gd} , and worker exit rate δ , while solving the initial BGP. On the other hand I estimate:

$$\Theta^{IRF} = \{\varepsilon, \eta, \nu, \phi, \theta, \lambda, \sigma\} \quad (29)$$

by matching empirical and model-generated impulse responses. Specifically, given a set of parameters, I compute the initial and final steady states, as well as the transition path induced by the exogenous series of regional birth rates. This produces a simulated dataset I then use to replicate the shift-share IV and estimate the SVAR-IV described in Section 2.

Clustering Commuting Zones Solving the theoretical model is time-consuming and the cost increases dramatically with the number of regions. Therefore I need to reduce this set from the 660 CZs used in the empirical part to estimate the structural parameters of the theory. To carry this out, I applied a K-means clustering procedure based on geographical coordinates and the variables in the VAR.

In particular, I first restrict the sample to CZs observed in all datasets required for the quantitative exercise. Then, given a number of clusters we want to produce, I form a 2010 cross-section and apply K-means to population-weighted latitude and longitude and to the annual changes in gross inflows, gross outflows, output per worker, average wages, employment relative to population, establishments per worker, and the housing-price index. This produces a set of clusters containing CZs that are similar in terms of migration, output and other economic variables, but that are also geographically close to each other. This rules out, for example, a cluster with some CZs in California as well as Florida.

To find the optimal number of clusters, I consider a set of potential partitions and compare their IRF coefficients with the ones in Section 3. Specifically, for each candidate partition, I reconstruct the cluster-level economic panel, migration shares, and birth-rate shift-share instrument. Once I re-estimate the “cluster-level” PVAR-IV, I can then compute its implied IRF $\hat{\mathbf{\Gamma}}^{cluster}$. I select the partition that minimizes:

$$\|\hat{\mathbf{\Gamma}}^{baseline} \oslash \hat{\mathbf{\Gamma}}^{cluster} - \mathbf{1}\| \quad (30)$$

where $\oslash$ denotes element-by-element division. The grouping is chosen to preserve the cumulative responses of gross inflows, gross outflows, output per worker, wages, and housing prices through year eight. The selected partition contains 29 model regions¹¹.

Exogenous Parameters I set the discount rate $\rho = 0.1$ as in [Crews \(2023\)](#), and target the average share of personal consumption expenditures on housing over the time period to set $\alpha = 0.818$. For the initial distribution of productivities $f(z)$, I use a truncated *Lognormal*(μ_f, σ_f) distribution using the estimates of the mean (92.3) and coefficient of variation (0.481) from [Huggett, Ventura and Yaron \(2006\)](#) to set μ_f and σ_f . To identify available land for housing T_g , I use raster data from the USGS National Land Cover as well as Data Elevation Models. Specifically, for each county I find the area (in square miles) that does not contain open water, wetlands or perennial snow as well as with an inclination lower than 15% following architectural guidelines and [Saiz \(2010\)](#). Then, for each model region, I construct T_g by summing the usable land measure across counties within each CZ, and later across CZs within each cluster.

I follow [Peters \(2022\)](#) parameterize bilateral trade costs as functions of distance by setting $\tau_{gd} = (d_{gd}/d_{min})^{\zeta_\tau}$, with $d_{gg} = d_{min}$. I compute each region’s geographic centroid as an initial-employment-weighted average of CZ coordinates and measure bilateral distance using the haversine formula. Furthermore, I set $d_{min} = 32\text{mi}$, which is the fifth percentile of the square root of CZ land area. I estimate ζ_τ by regressing bilateral prices on distance, similar to [Castro-Vincenzi et al. \(2024\)](#).

In particular, I use the Commodity Flow Survey (CFS) datasets from 2012 and 2017, which provide information on domestic freight shipping within the US. For each shipment i , I observe its weight and value, as well as the great circle distance d_i between shipment origin and destination. Thus, since in the model $p_{gd}(t) \propto \tau_{gd}$, I estimate the following regression:

$$\log p_{i,t} = \zeta_\tau \log(d_i) + \beta_s \mathbf{1}\{o = d\} + \mathbf{X}_{i,t} + \epsilon_{i,t} \quad (31)$$

where $p_{i,t}$ is the value per pound of the shipment and $\mathbf{X}_{i,t}$ are a set of controls such as destination d , origin o and quarter fixed effects. The estimates are reported in Table 5 in the Appendix and I use the estimate in the column (4).

For the return-to-skill shifter χ_g , I use Quarterly Workforce Indicators data. This source provides information on earnings and employment by region, year and four education groups. Therefore, I map the latter categories into ordered quantile intervals of the truncated skill distribution. For each education group k , construct a centered skill score r_k , which I use to

¹¹I evaluate partitions containing between 10 and 29 regions. I also consider 2 K-means initialization seeds, and use initialization seed 1. To aggregate each variable at the cluster level, I sum flow and stock quantities across CZs within each region and reconstruct ratios from the aggregated totals. I also use aggregated bilateral migration flows and origin birth rates to build the shift-share instrument. Region coordinates are employment-weighted CZ centroids, which I use to compute bilateral great-circle distances.

estimate:

$$\log \bar{w}_{gkt} = \delta_{gt} + \zeta_{kt} + \zeta_E r_k l_{gt} + \chi_g r_k + u_{gkt} \quad (32)$$

where l_{gt} is log regional employment relative to the regional mean. I normalize the estimates so that $\sum_g \omega_g \chi_g = 0$, where ω_g is the region's average employment share.

Finally, I assume the economy is at a BGP in 2002, the first year in my sample, and set γ_L to 1.25% by targeting the average growth rate of the civilian labor force in the 10 years preceding my sample. For total population I use the sum of wage employment in that year from the sample in Section 2 so that $\bar{N} \approx 134$ mil. workers.

Internal Estimation For a given set Θ^{IRF} , I estimate the remaining regional fundamentals by jointly solving the initial steady state and targeting moments from the cross-section in 2002, the first year in the sample. Productivities A_g matches real output per worker whereas amenities D_g match population shares. I also calibrate a symmetric matrix of bilateral migration shifters κ_{gd} to the origin-destination migration flows in the data, while setting $\kappa_{gg} = 1$. I use a modified version of Algorithm B where I solve for the fixed point of parameters in addition to endogenous variables. This also allows me to find δ by using equation (28), the calibrated population growth rate γ_L and the birth rates $\bar{b}_g^1$ observed 25 years before the first year in my sample. These are the initial elements in the series $\{b_g(t)\}$ I will use to simulate the transition. These internally calibrated objects are recomputed for every candidate structural vector because their mapping into the cross-sectional targets depends on the common structural parameters in Θ^{IRF} . Once this is complete, I compute the second endpoint for the transition. To solve the final BGP, I use the same common parameters and regional fundamentals. The only difference with the initial steady state are the lagged birth rates, which now correspond to those in the last year of my sample $\bar{b}_g^2$.

With both endpoints in hand, I can now use Algorithm B to compute the transition dynamics of the economy. In particular, I leave all parameters constant except birth rates $\{b_g(t)\}$, which are the same as the ones I use to build the shift-share instrument. This results in a panel series of 17 years (including the first and last ones that I use for the BGP computation). Note it is unlikely the economy reaches the second steady state after this period. Therefore, I assume $b_g(t) = \bar{b}_g^2$ for all t thereafter and set the total periods to a higher number ($T = 150$). After solving the transition, I aggregate the continuous-time solution to annual frequency and retain only the years corresponding to the empirical panel in the IRF estimation. I also use the annual model solution to reconstruct a simulated shift-share instrument. I use the model bilateral migration flows to build lagged origin-destination exposure shares and interact those shares with the empirical origin birth-rate series. I then estimate the destination-region PVAR-IV on the simulated panel.

To estimate Θ^{IRF} , I follow an impulse response matching procedure that targets migration inflows, output per worker, and housing prices. Specifically, I repeat the internal

estimation steps described above for a sequence of parameters in a Sobol grid and seek to minimize the following objective function:

$$J = \min_{\Theta^{IRF}} \sum_j \sum_{h=0}^4 \left[\frac{\Gamma_{jh}(\Theta)}{\hat{\Gamma}_{jh}} - 1 \right]^2 \quad (33)$$

where $\hat{\Gamma}_{jh}$ is the cumulative response of variable j in horizon h , estimated using the 35 clusters, and $\Gamma_{jh}(\Theta^{IRF})$ if the equivalent for the model-generated response. The targeted responses discipline endogenous migration adjustment, agglomeration-driven productivity, and housing-related congestion, the three central forces driving the relevant dynamics in the quantitative model. I omit the response of gross inflows for $h = 0$ since this is normalized to one.

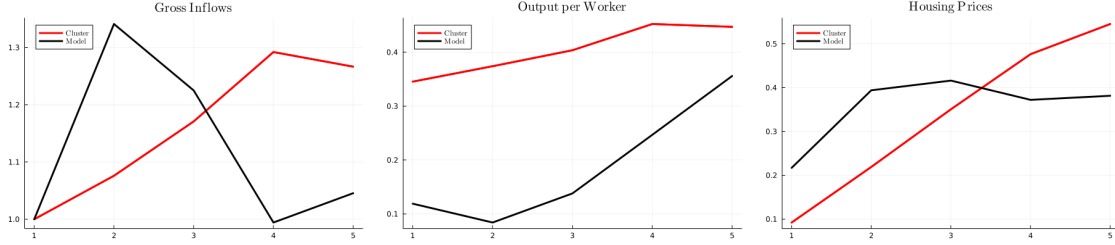
5.2 Parameter Estimates

Table 4 summarizes all the parameter values, including those that are calibrated externally, calibrated while solving the initial BGP, as well as the ones that minimize the IRF distance. The seven free parameters are estimated jointly from 14 cumulative responses for gross inflows, output per worker, and housing prices. Furthermore, Figure 7 shows the model fit of the targeted impulse responses. It compares the cumulative responses from the selected CZ aggregation with the responses obtained after simulating the model at the minimum-distance parameter vector.

TABLE 4 STRUCTURAL PARAMETERS

External Calibration				
<i>Parameter</i>	<i>Description</i>	<i>Mean/Value</i>	<i>Range</i>	<i>Source</i>
ρ	Discount Parameter	0.10	-	Crews (2022)
α	Consumption Share	0.82	-	Personal Consumption Expenditures
$\bar{z}^f$	Mean of Initial Distribution	92.3	-	Hugget (2006)
CV^f	Coeff. of Variation of Initial Distribution	0.481	-	Hugget (2006)
T_g	Regional Land	79326.60	[16294.91 ; 206446.18]	USGS geodata
χ_g	Regional Return-to-Skill	-0.09	[-0.56 ; 0.19]	QWI
Internal Calibration				
<i>Parameter</i>	<i>Description</i>	<i>Mean/Estimate</i>	<i>Range</i>	<i>Moment</i>
δ	Worker Exit Rate	1.94E-04	-	Population Growth in BGP 1
A_g	Regional Productivity	2.51E-04	[2.06E-04 ; 3.48E-04]	Output per Worker in BGP 1
D_g	Regional Amenities	16.80	[0.51 ; 51.18]	Population Shares in BGP 1
Impulse Response Matching				
<i>Parameter</i>	<i>Description</i>	<i>Estimate</i>	<i>Moment</i>	
ε	Elasticity of Substitution Across Varieties	3.82	Impulse Responses to Migration Inflow Shock	
σ	Elasticity of Substitution Across Varieties	0.55	Impulse Responses to Migration Inflow Shock	
η	Inter-temporal Elasticity	0.24	Impulse Responses to Migration Inflow Shock	
ν	Migration Elasticity	4.02	Impulse Responses to Migration Inflow Shock	
λ	Migration Opportunity Rate	0.02	Impulse Responses to Migration Inflow Shock	
ϕ	Elasticity of Amenities to Population	1.63	Impulse Responses to Migration Inflow Shock	
θ	Share of Land in Housing	0.91	Impulse Responses to Migration Inflow Shock	

FIGURE 7. MODEL FIT - TARGET IRFs



Note: The red line represents the cumulative impulse response (CIRF) estimated from the 29 K-means clusters. The black line is the CIRF estimated by simulating the model with the selected Sobol parameter vector. The three panels – gross inflows, output per worker, and housing prices – are the IRFs entering the minimum-distance criterion.

6 Fertility, Removals and Relocation

How does the economy respond over time to population shocks? Although the IRFs in Section 3 show what the effect of migration flows on local conditions is, we cannot determine the impact of shocks to the stock of workers, such as sudden removals. Furthermore, the estimates are not enough to infer the general equilibrium effects that would follow after demographic changes nor the relevance of population flows on aggregate variables. Therefore, to answer these questions, I use the calibrated model. Specifically, I consider three exercises separately.

The first explores how the economy reacts to the post-estimation path of observed birth rates. The second is a labor-removal (“deportation”) shock that decreases the total number of workers in the economy by reducing the population in certain areas while keeping the size of the remaining regions constant. The third one is a “relocation” shock: instead of maintaining the population of other locations, I increase it by a uniform share so as to keep the total number of households in the economy constant.

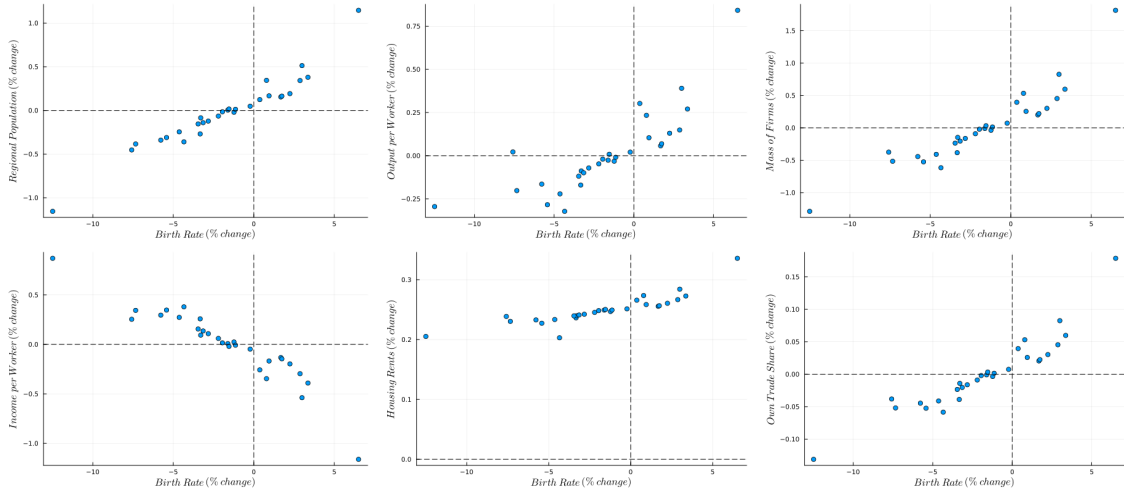
An important element of these counterfactuals is that these local shocks are a geographically heterogeneous. In the first exercise, birth rates differ by locality whereas in the other two the local population share that is removed or relocated differs across treated regions. To calibrate these shares I use the estimates from [Passel and Krogstad \(2023\)](#) to identify the states with the highest number of undocumented immigrants. These include California, Texas, Florida and Illinois. I assign a state-specific shock to a model region when more than 30 percent of its constituent counties are located in that state. Consequently, we can think of the second shock as a general deportation of approximately 1% of the labor force. By region however, this percentage varies from 3.2% to 5.4%. On the other hand, in the third exercise these workers are redistributed to places with a lower incidence of undocumented immigration, in a similar manner to the transportation of migrants to sanctuary cities by some states in 2023 (see for example [Goodman et al. \(2024\)](#)).

6.1 New Birth Rates

To estimate the model, I used regional birth rates lagged by 25 years and simulated the transition for 18 years, assuming the economy starts at a balanced growth path in 2000. I now carry out a similar exercise, beginning a new transition in the year 2019, with lagged regional birth rates from the 1994-2000 period. After 2025, I hold regional entry rates constant to define the terminal balanced growth path. Figures 8 and 9 report the adjustment through 2041 rather than the economy's fully converged terminal allocation.

On average fertility in this time experienced a small drop of -0.05 percentage points by 1997, followed by a increase that left the total drop in the period at -0.02 percentage points. For reference, this implies the population growth rate along the BGP drops from 1.53% to 1.5% . However, even a slight decrease such as this one can generate interesting general equilibrium dynamics across space.

FIGURE 8. LONG-TERM CHANGE AFTER NEW BIRTHS



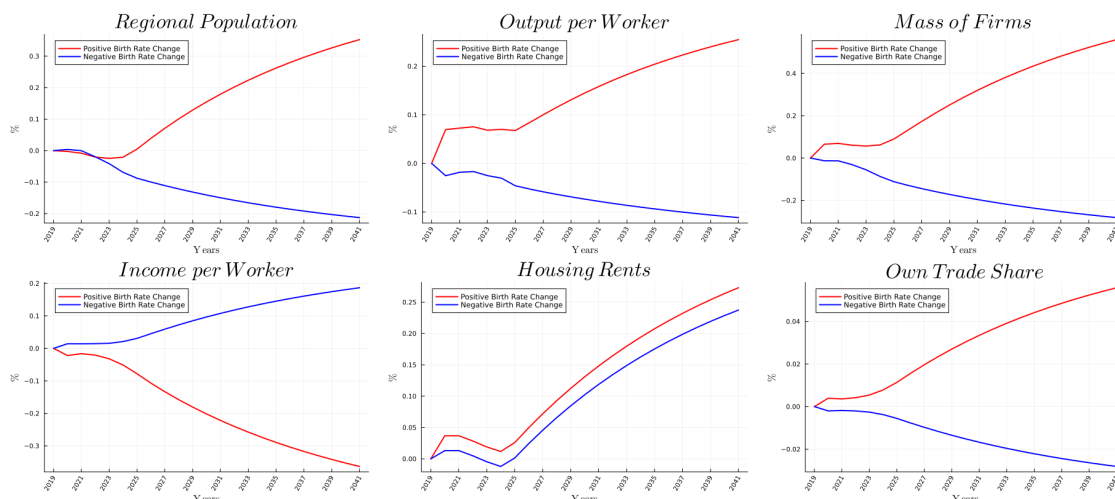
Note: Each panel plots the percentage change in the indicated regional outcome between 2019 and 2041 against the percentage change in the region's birth rate over the same horizon. Each marker represents one model region, and the dashed lines mark zero changes.

Figure 8 plots the total birth rate change in a region between 2019 and 2041, along with that of several local outcomes. First, we see that areas with higher birth rates experience a rise in their population. Due to migratory patterns, even some locations with a decrease in fertility have a higher population in 2041 than in 2019. However, since the value of λ implies high migration frictions, most of the increase in population comes from the rise in birth rates rather than net-inflows from other places. This increase in population leads to a higher mass of firms in the long run, which in turn raises output per worker in these regions.

Nevertheless, this rise in productivity is also accompanied by a drop of real income per worker. Recall this measure corresponds to the regional wage divided by a price index that combines housing rents and the price of tradable goods. On the one hand, a higher number

of firms makes the region more competitive. Furthermore, housing rents rise since the high share of scarce land in housing production implies the congestion force dominates the higher supply of labor. Furthermore, these negative effects on income per worker are amplified by the direct negative effect of a higher own-trade share, which raises prices of the tradable goods produced locally. This leads to a lower real income in locations with higher birth rates.

FIGURE 9. CUMULATIVE CHANGE BY TYPE OF SHOCK



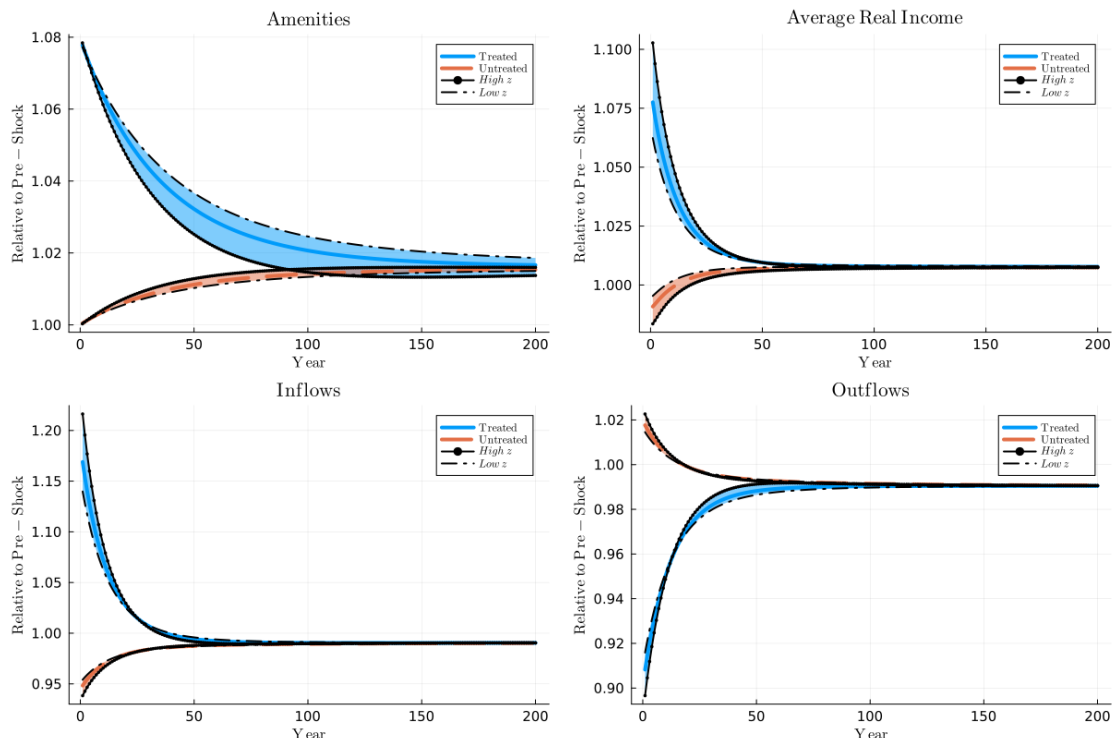
Note: Each panel reports the unweighted mean percentage deviation from the region's 2019 value for regions whose birth rate is higher in 2041 (red) and regions whose birth rate is lower in 2041 (blue). The paths are changes relative to 2019, not cumulative sums of annual growth rates.

How do these variables evolve over time? While Figure 8 shows the long-term effect, Figure 9 plots the cumulative responses as birth-rates transition to the new steady state. These effects are grouped by whether regions experienced a positive and negative long-term change in birth-rates, respectively. For reference, the former group experienced on average a 0.03 *p.p.* increase in fertility, whereas the latter saw their birth-rates drop by 0.06 *p.p.* Regions with higher births exhibit a faster change in population than the former, which start around 2021. This is due to the fact that both groups experience an initial fall in fertility, although one of them has higher birth rates once it reaches the second balanced growth path.

The long-term increase in population is higher in magnitude than the drop in regions with decreasing fertility. Note these values imply a 1 *p.p.* rise of fertility increases population by approximately 10%, whereas a similar drop decreases population by 3.3%, revealing asymmetric effects when accounting for general equilibrium migration flows. We can observe this asymmetry in the rest of the plots in Figure 9, as measures of productivity and household welfare follow similar dynamics to those of population. Indeed, regions with higher birth-rates see an initial 0.08% rise in output per worker followed by a continued

rise to 0.25%. This is driven by a mass of firms that is 0.5% higher. Nonetheless, real incomes fall continuously when birth-rates rise. This is due to the mechanisms we explored above, where a higher own-trade share and demand for housing services increase the prices households face in these regions, which dominates the gains in productivity.

FIGURE 10. DEPORTATION SHOCK: UTILITY AND POPULATION FLOWS

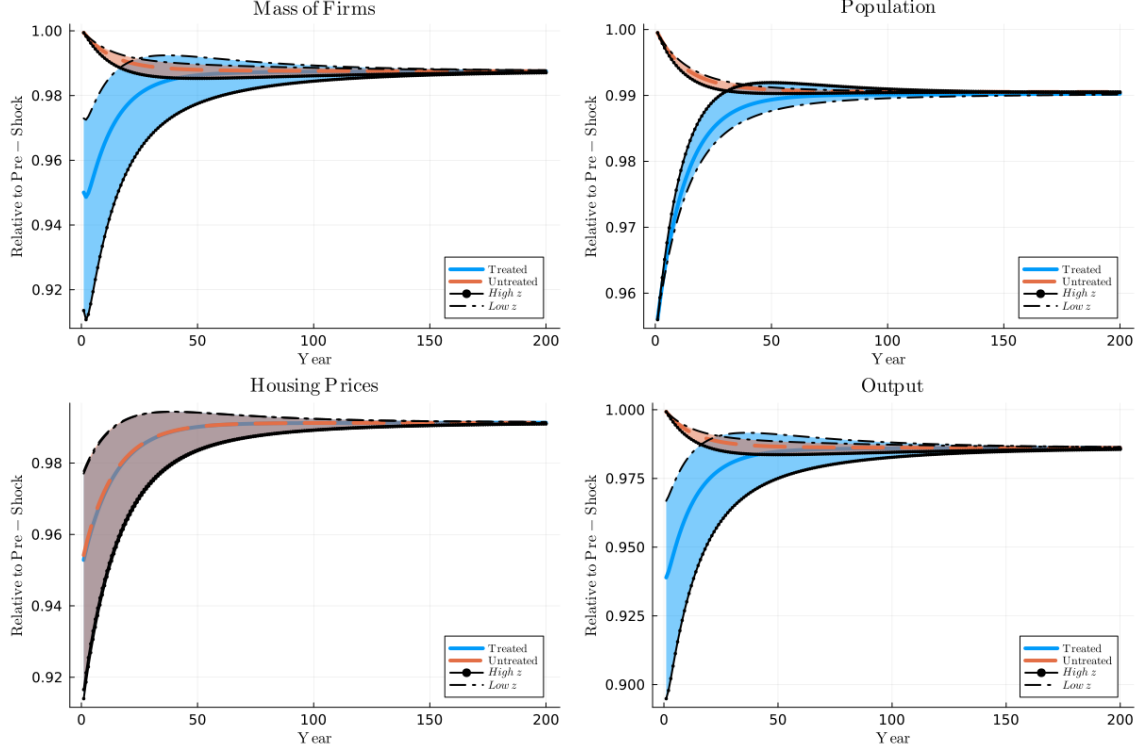


Note: Responses of amenities, real income, and gross migration inflows to the “deportation” shock where heterogeneous regional population shares are removed from the economy. The blue line represents the response in regions that experience a population reduction (“treated”) whereas the red line represents the effect in the remaining ones (“untreated”). In both settings, the policy removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Variables x_t are plotted relative to their pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

These results are particularly informative for recent debates about the implications of declining birth rates. On the one hand, the lower population implied by reduced fertility could improve real incomes per worker, and this could partly offset the initial decline in fertility by making childbearing more affordable for households. On the other hand, population decline may also reduce the availability of local services, amenities, and employment opportunities. This deterioration in local attractiveness could induce workers to migrate toward “sponge cities” that absorb households from declining regions, thereby raising prices in those destinations, as the impulse responses in Section 3 suggest. The counterfactual responses in Figure 9, however, indicate that the effects of future declines in

birth rates are likely to be more nuanced. Regions with lower fertility experience declines in productivity and in the number of firms, but they also increasingly rely on tradable goods produced elsewhere. As a result, these regions can become more attractive despite weaker local production, because lower local costs raise real incomes.

FIGURE 11. DEPORTATION SHOCK: POPULATION, VARIETIES AND OUTPUT



Note: Responses of varieties, population, housing prices and output to the “deportation” shock where heterogeneous regional population shares are removed from the economy. The blue line represents the response in regions that experience a population reduction (“treated”) whereas the red line represents the effect in the remaining ones (“untreated”). In both settings, the policy removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Variables x_t are plotted relative to their pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

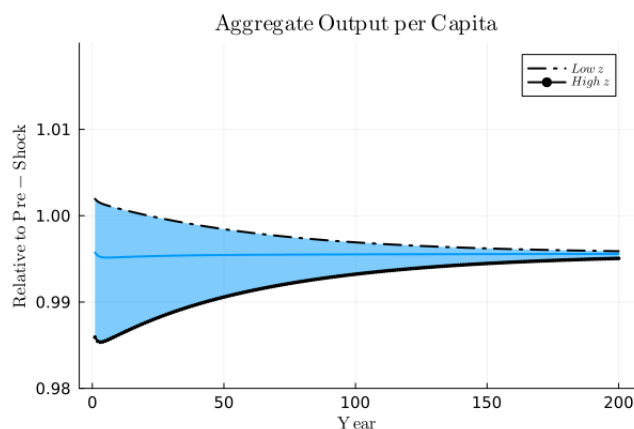
6.2 Labor Removal and Relocation

Labor Removal The labor-removal counterfactual consist of removing the state-specific shares described above in affected regions, while maintaining the population of all other regions unchanged. On the aggregate, this means national population falls and the economy starts to converge to a new BGP. To analyze how skill composition affects the results, I compare a proportional removal across the full skill distribution with low- z and high- z removal scenarios. Figure 10 displays the effects of the initial shock. Remaining residents in regions directly impacted by the deportation experience less congested amenities and

an average increase in real incomes of approximately 7–8%, after adjusting for prices of consumption goods and housing services. As these regions become more attractive, more workers choose to move there, resulting in an immediate 20% increase in gross inflows and a 5% decrease in outflows, while the opposite trends are observed in non-treated areas. Over time, as households choose to leave these non-treated regions, their amenities also become less congested.

Over time, the effects of the initial shock begin to dissipate, and most variables stabilize at new levels relative to the original steady state. With fewer workers in the economy, amenities in the long run increase by 2%, and real incomes rise by 1%. This aligns with the long-term effect on output, shown in Figure 11, which exhibits a decrease of more than 5% in affected areas. Other regions initially see a muted response in production, but they eventually also lose population as migration reallocates workers toward targeted locations. This also helps explain why housing prices also drop in these areas: given the high share of land in housing production, a drop in demand from higher outflows will lead to lower prices. Notably, if the shock were to target high-skilled individuals instead of being uniformly distributed across the regional productivity distribution $\psi_g(z, t)$, output would experience a larger decline, while real incomes would be higher only in targeted areas. Intuitively, such a policy constitutes a labor supply shock that lowers the average skill level available in affected regions, thereby increasing lowering total output.

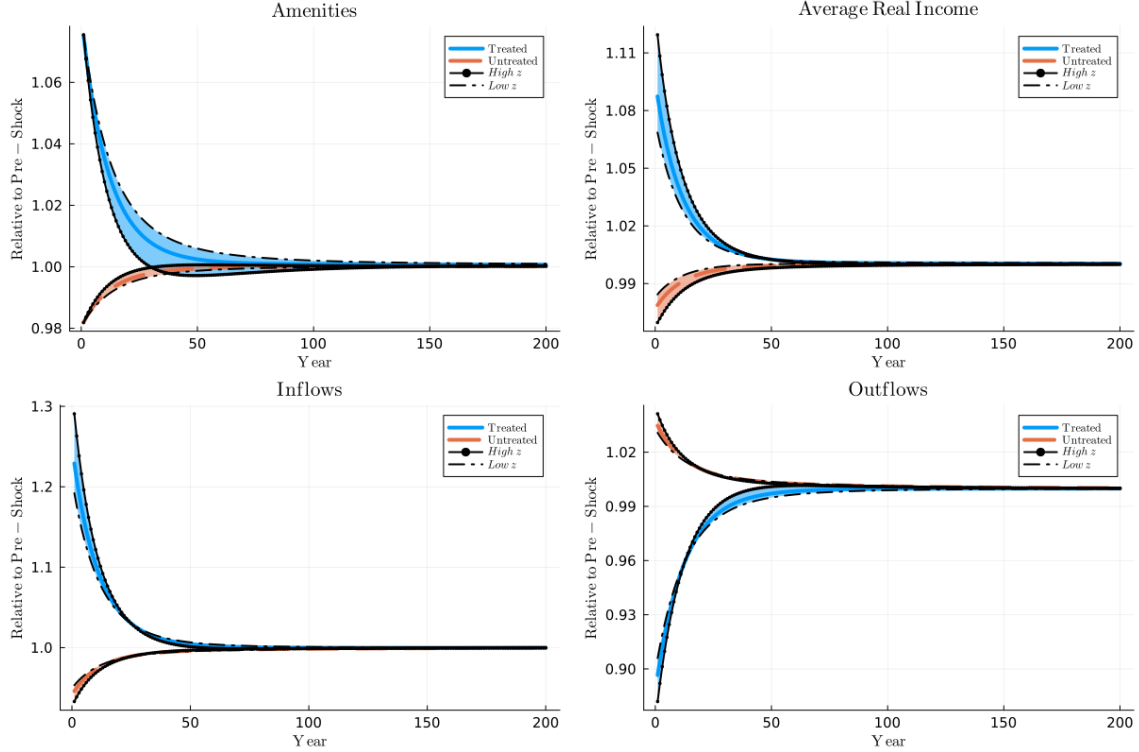
FIGURE 12. DEPORTATION SHOCK: AGGREGATE OUTPUT



Note: Response of aggregate output per worker to the “deportation” shock where heterogeneous regional population shares are removed from the economy. The blue line represents the response when the shock removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Output per worker is plotted relative to its pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

Local populations also converge to a new steady state that is 1% lower than the level observed before the policy was implemented. A smaller workforce implies reduced migration flows, as shown in Figure 10. Additionally, a lower labor supply means fewer firms

FIGURE 13. RELOCATION SHOCK: UTILITY AND POPULATION FLOWS

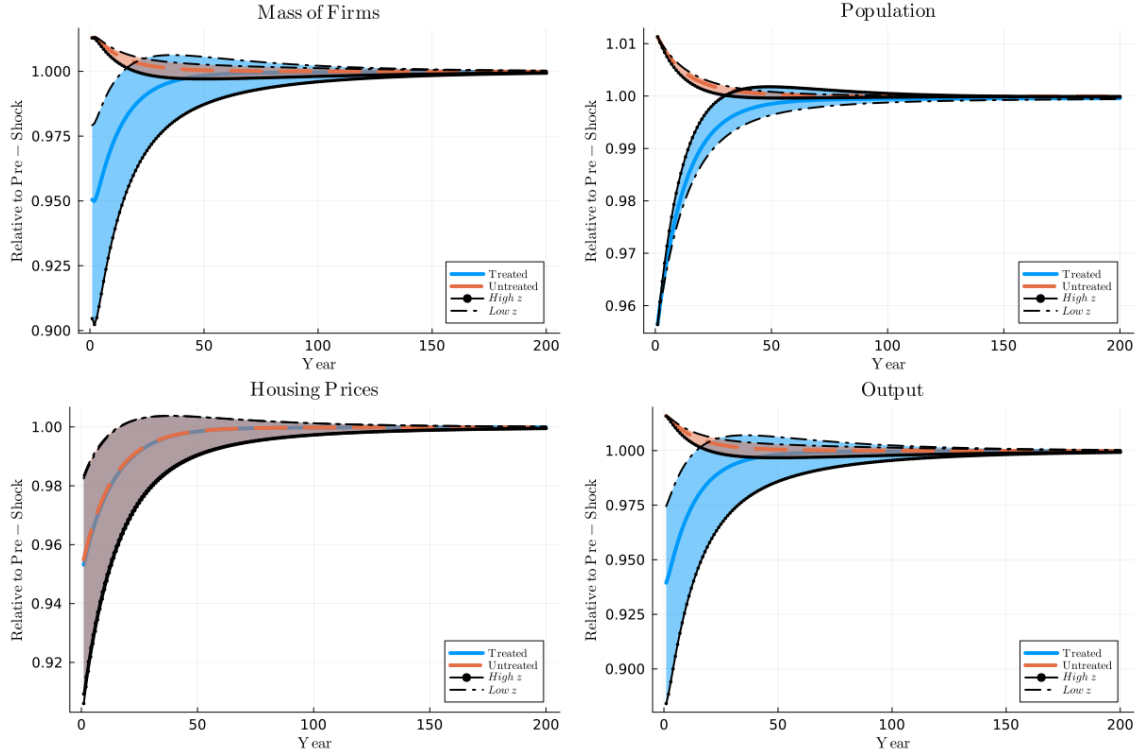


Note: Responses of amenities, real income, and gross migration inflows to the “relocation” shock where heterogeneous regional population shares are moved from treated to untreated areas. The blue line represents the response in regions that experience a population reduction (“treated”) whereas the red line represents the effect in the receiving ones (“untreated”). In both settings, the policy removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Variables x_t are plotted relative to their pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

can cover the fixed costs of operation, leading to a decrease in the number of available varieties. This effect is further amplified by the dynamic influence of $M_g(t)$ on fixed costs, as described in equation (12). Together, these mechanisms contribute to a steady decline in the mass of firms, illustrated in Figure 11. A similar process unfolds in unaffected areas, where population loss due to migration outflows also reduces labor and variety levels, ultimately decreasing output in those regions. Meanwhile, in the regions affected by the shock, migration inflows gradually increase the variety of goods over time.

In terms of aggregate effects, the deportation shock results in a long-run decline in output per worker of 0.5%, as shown in Figure 12. This new steady state arises because workers exit local economies without being replaced elsewhere, unlike the “relocation” shock I explore later. Consequently, the overall economy is left with a smaller labor supply and, therefore, fewer varieties, which reduces productivity. This outcome is similar to an inverse version of the refugee allocation scenario in Peters (2022), where output per worker also rises

FIGURE 14. RELOCATION SHOCK: POPULATION, VARIETIES AND OUTPUT

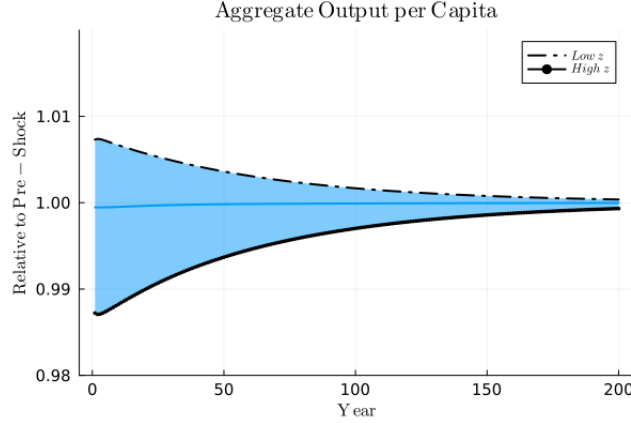


Note: Responses of varieties, population, housing prices and output to the “relocation” shock where heterogeneous regional population shares are moved from treated to untreated areas. The blue line represents the response in regions that experience a population reduction (“treated”) whereas the red line represents the effect in the receiving ones (“untreated”). In both settings, the policy removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Variables x_t are plotted relative to their pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

in the long run. However, in this case, the effect is driven by agglomeration externalities across all markets, rather than being limited to the manufacturing sector. Additionally, it is worth noting the different impacts depending on the type of workers removed. If high-skilled workers leave, aggregate output per capita immediately decreases to the new level. By contrast, if the shock removes low-skilled labor, output per capita initially rises, as the average skill level in the economy increases. Over time, however, the same dynamics that reduce varieties come into play, though at a slower rate.

Relocation How would aggregate productivity react if instead of removing workers they were relocated them to other regions? I distribute the displaced workers across all remaining regions through a common proportional increase chosen to keep national population fixed. Effectively, this policy also takes the economy out of its initial steady state but does not change the BGP towards which the economy transitions to in the long-run. Figure 13

FIGURE 15. RELOCATION SHOCK: AGGREGATE OUTPUT



Note: Response of aggregate output per worker to the “deportation” shock where heterogeneous regional population shares are removed from the economy. The blue line represents the response when the shock removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Output per worker is plotted relative to its pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

shows that, similar to before, regions that lose their population see an increase of inflows and a reduction of outflows, whereas the opposite is true of areas that receive the initial influx. Right after the shock, congested amenities and lower real incomes drive migrants back to areas whose population was relocated. This dynamic continues for 50 years until the number of workers in each places converges back.

The mass of firms in each region also transition back to the initial steady state, as well as output. In source regions, proportional relocation lowers firm mass and total output by more than 5% at the beginning of the transition, while relocation of the workers in the top 10% of the local skill distribution generates declines close to 10%. However, unlike population, these variables take longer to converge. The number of producers is a slow moving variable given that the fixed cost of production is a state variable that evolves gradually following equation (12). It therefore takes time to adjust, which is why the initial drop in shocked areas also occurs in a longer time frame relative to labor.

Finally, Figure 15 shows what the aggregate effects of this relocation policy are. Although in the short- and long-run, proportional relocation leaves output per capita close to the pre-shock level, high-skill worker relocation generates an initial drop of approximately 1%. Displacing low-skill workers has an almost symmetrical effect, increasing output per worker by a similar magnitude. Although the paths converge back to one, it is worth noting how changing the location of households across space, without changing fundamental parameters of the economy, can produce significant effects that last for several years.

7 Conclusion

This paper studies the dynamic effect of increasing population inflows on local markets. Most of the previous research on labor mobility and its impact on economic outcomes focused on either short or long-term responses, but did not consider transitional dynamics. Although some papers address this element, their identification of structural migration shocks relies on timing assumptions, whereas the analysis above adopts an SVAR-IV approach using instruments based on weather variations. The results reveal labor productivity, wages and employment opportunities in urban economies respond positively to a 1% increase in population inflows.

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A Theory Appendix

This appendix collects the derivations behind the equilibrium conditions in the main text. Workers supply effective labor inelastically. A worker with raw skill z supplies

$$a_g(z) = z \exp\{\chi_g(\log z - \mu_z)\}, \quad \mu_z = \int_{\Omega_z} \log z f(z) dz, \quad (34)$$

in region g . The benchmark $\chi_g = 0$ implies $a_g(z) = z$.

A.1 Static equilibrium

A.1.1 Households

A worker of type z residing in g earns $a_g(z)W_g(t)$ and chooses consumption and housing according to

$$\max_{C_g, H_g} D_g(t) C_g^\alpha H_g^{1-\alpha} \quad \text{subject to} \quad P_g(t) C_g + P_g^h(t) H_g = a_g(z) W_g(t). \quad (35)$$

The expenditure rules and indirect flow utility are

$$C_g(z, t) = \frac{\alpha a_g(z) W_g(t)}{P_g(t)}, \quad H_g(z, t) = \frac{(1 - \alpha) a_g(z) W_g(t)}{P_g^h(t)}, \quad (36)$$

$$U_g(z, t) = \frac{D_g(t) a_g(z) W_g(t)}{P_g(t)^\alpha P_g^h(t)^{1-\alpha}}. \quad (37)$$

The CES aggregator over regional goods implies

$$c_{rg}(z, t) = \left(\frac{p_{rg}(t)}{P_g(t)} \right)^{-\sigma} C_g(z, t), \quad P_g(t) = \left(\sum_{r \in G} p_{rg}(t)^{1-\sigma} \right)^{1/(1-\sigma)}. \quad (38)$$

A.1.2 Housing

Competitive housing producers use land and labor according to

$$H_g(t) = \Gamma_h T_g^\theta L_g^h(t)^{1-\theta}, \quad \Gamma_h = \theta^{-\theta} (1 - \theta)^{-(1-\theta)}. \quad (39)$$

Cost minimization and zero profits imply

$$P_g^h(t) = W_g(t)^{1-\theta} R_g(t)^\theta. \quad (40)$$

Combining this expression with land-market clearing gives the housing supply curve

$$H_g(t) = \left(\frac{P_g^h(t)}{W_g(t)} \right)^{(1-\theta)/\theta} \frac{T_g}{\theta}. \quad (41)$$

Aggregate housing demand follows by integrating (36). Equating demand and supply yields

$$P_g^h(t) = W_g(t) \left(\frac{(1-\alpha)\theta L_g(t)}{T_g} \right)^\theta, \quad (42)$$

where total effective labor in region g is

$$L_g(t) = N(t) \int_{\Omega_z} a_g(z) \psi_g(z, t) dz. \quad (43)$$

The housing sector employs the constant share

$$L_g^h(t) = (1-\theta)(1-\alpha)L_g(t), \quad (44)$$

and land rents satisfy $R_g(t)T_g = \theta(1-\alpha)W_g(t)L_g(t)$.

A.1.3 Tradable goods and intermediate producers

The competitive producer of the variety shipped from g to d combines intermediate inputs as

$$y_{gd}(t) = \frac{1}{\tau_{gd}} \left(\int_0^{M_g(t)} y_{gd}(j, t)^{(\varepsilon-1)/\varepsilon} dj \right)^{\varepsilon/(\varepsilon-1)}. \quad (45)$$

Cost minimization gives

$$p_{gd}(t) = \tau_{gd} Q_g(t), \quad Q_g(t) = \left(\int_0^{M_g(t)} q_g(j, t)^{1-\varepsilon} dj \right)^{1/(1-\varepsilon)}. \quad (46)$$

An intermediate producer has labor requirement

$$l_g(j, t) = \frac{y_g(j, t)}{A_g} + F_g(t). \quad (47)$$

Monopolistic competition and free entry imply, respectively,

$$q_g(j, t) = \frac{\varepsilon}{\varepsilon-1} \frac{W_g(t)}{A_g}, \quad (48)$$

$$y_g(j, t) = (\varepsilon-1)A_g F_g(t), \quad l_g(j, t) = \varepsilon F_g(t). \quad (49)$$

All intermediate firms in a region therefore choose the same price and scale. It follows that

$$Q_g(t) = M_g(t)^{1/(1-\varepsilon)} \frac{\varepsilon}{\varepsilon-1} \frac{W_g(t)}{A_g}, \quad (50)$$

$$L_g^y(t) = \varepsilon M_g(t) F_g(t). \quad (51)$$

Using (44), labor-market clearing implies

$$L_g^y(t) = c_L L_g(t), \quad c_L \equiv \alpha + \theta(1-\alpha). \quad (52)$$

The share of destination d 's expenditure allocated to origin g is

$$\pi_{gd}(t) = \frac{[\tau_{gd} M_g(t)^{1/(1-\varepsilon)} W_g(t)/A_g]^{1-\sigma}}{\sum_{i \in G} [\tau_{id} M_i(t)^{1/(1-\varepsilon)} W_i(t)/A_i]^{1-\sigma}}. \quad (53)$$

Total goods expenditure is $c_L W_g(t) L_g(t)$. Goods-market clearing then reduces to the balanced-trade system

$$W_g(t) L_g(t) = \sum_{d \in G} \pi_{gd}(t) W_d(t) L_d(t), \quad g \in G, \quad (54)$$

with one wage used as the numeraire.

The domestic trade share can be used to express the consumption price index as

$$P_g(t) = \frac{\varepsilon}{\varepsilon-1} \pi_{gg}(t)^{1/(\sigma-1)} M_g(t)^{1/(1-\varepsilon)} \frac{W_g(t)}{A_g}. \quad (55)$$

Substituting (42) and (55) into (37) gives the flow utility used in the dynamic problem:

$$U_g(z, t) = D_g(t) a_g(z) \frac{\Gamma_g}{\Gamma_U} \pi_{gg}(t)^{\alpha/(1-\sigma)} M_g(t)^{\alpha/(\varepsilon-1)} L_g(t)^{-\theta(1-\alpha)}, \quad (56)$$

where

$$\Gamma_U = \left(\frac{\varepsilon}{\varepsilon-1} \right)^\alpha ((1-\alpha)\theta)^{\theta(1-\alpha)}, \quad \Gamma_g = A_g^\alpha T_g^{\theta(1-\alpha)}. \quad (57)$$

For completeness, regional output can be written as

$$Y_g(t) = A_g \frac{\varepsilon-1}{\varepsilon} L_g^y(t) M_g(t)^{1/(\varepsilon-1)}. \quad (58)$$

Equation (58) makes the variety channel transparent: holding production labor fixed, a larger mass of firms raises regional output.

A.1.4 Evolution of the fixed cost

The fixed labor cost follows

$$\frac{\dot{F}_g(t)}{F_g(t)} = \log \left(\frac{F_E}{F_g(t) M_g(t)^\eta} \right). \quad (59)$$

Combining (59) with (51) and (52) gives a law of motion stated only in terms of $F_g(t)$ and $L_g(t)$:

$$\frac{\dot{F}_g(t)}{F_g(t)} = \log \left[\frac{F_E \varepsilon^\eta F_g(t)^{\eta-1}}{c_L^\eta L_g(t)^\eta} \right]. \quad (60)$$

A.2 Dynamic equilibrium

A.2.1 Migration choice and the HJB equation

Migration opportunities arrive at Poisson rate λ . Conditional on an opportunity, a worker currently in g draws independent Fréchet preference shocks ξ_d and chooses the destination that maximizes

$$\xi_d \kappa_{gd} V_d(z, t), \quad (61)$$

where $\kappa_{gd} \leq 1$ is the bilateral mobility shifter and $\kappa_{gg} = 1$. The Fréchet scale is normalized so that the expected shock equals one. Standard extreme-value calculations imply

$$\mathcal{I}_g(z, t) \equiv \mathbb{E}_\xi \left[\max_{d \in G} \{ \xi_d \kappa_{gd} V_d(z, t) \} \right] = \left[\sum_{d \in G} (\kappa_{gd} V_d(z, t))^\nu \right]^{1/\nu}, \quad (62)$$

$$m_{gd}(z, t) = \frac{(\kappa_{gd} V_d(z, t))^\nu}{\sum_{i \in G} (\kappa_{gi} V_i(z, t))^\nu}. \quad (63)$$

Because $a_g(z)$ is non-separable, $V_g(z, t)$ and hence $m_{gd}(z, t)$ generally vary with raw skill.

Over a short interval of length Δ , the Bellman equation is

$$V_g(z, t) = \Delta U_g(z, t) + (1 - \rho\Delta)(1 - \delta\Delta) \left[(1 - \lambda\Delta) V_g(z, t + \Delta) + \lambda \Delta \mathcal{I}_g(z, t + \Delta) \right] + o(\Delta). \quad (64)$$

Subtracting $V_g(z, t)$, dividing by Δ , and taking $\Delta \rightarrow 0$ gives

$$(\delta + \rho + \lambda) V_g(z, t) = U_g(z, t) + \partial_t V_g(z, t) + \lambda \mathcal{I}_g(z, t). \quad (65)$$

To obtain the representation used in the main text, define

$$\delta_{gd}(z, t) \equiv \frac{\kappa_{gd}}{G} m_{gd}(z, t)^{-(1+\nu)/\nu}. \quad (66)$$

Equations (62) and (63) imply

$$\mathcal{I}_g(z, t) = \sum_{d \in G} m_{gd}(z, t) \delta_{gd}(z, t) V_d(z, t). \quad (67)$$

Using $\sum_d m_{gd}(z, t) = 1$, equation (65) becomes

$$(\delta + \rho)V_g(z, t) = U_g(z, t) + \partial_t V_g(z, t) + \lambda \sum_{d \in G} m_{gd}(z, t) [\delta_{gd}(z, t) V_d(z, t) - V_g(z, t)]. \quad (68)$$

A.2.2 Kolmogorov forward equation

Let $\Psi_g(z, t) = \Pr\{Z \leq z, g(t) = g\}$ and let $\psi_g(z, t)$ denote its density with respect to z . The number of workers in region g with raw skill no greater than z is $N(t)\Psi_g(z, t)$. Over an interval Δ ,

$$\begin{aligned} N(t + \Delta)\Psi_g(z, t + \Delta) &= \Delta E_g(t)F(z) + (1 - \delta\Delta) \left[(1 - \lambda\Delta)N(t)\Psi_g(z, t) \right. \\ &\quad \left. + \lambda\Delta \sum_{i \in G} \int_{-\infty}^z m_{ig}(s, t)N(t)\psi_i(s, t) ds \right] + o(\Delta). \end{aligned} \quad (69)$$

Subtracting the initial stock, dividing by Δ , taking the limit, and differentiating with respect to z yields

$$\begin{aligned} \partial_t \psi_g(z, t) &= \frac{E_g(t)}{N(t)} f(z) - \delta \psi_g(z, t) - \lambda \left[(1 - m_{gg}(z, t)) \psi_g(z, t) - \sum_{i \neq g} m_{ig}(z, t) \psi_i(z, t) \right] \\ &\quad - \frac{\dot{N}(t)}{N(t)} \psi_g(z, t). \end{aligned} \quad (70)$$

Integrating over skill gives regional population,

$$N_g(t) = N(t) \int_{\Omega_z} \psi_g(z, t) dz, \quad (71)$$

while summing across regions gives aggregate population dynamics,

$$\dot{N}(t) = \sum_{g \in G} E_g(t) - \delta N(t). \quad (72)$$

A.3 Detrending and the balanced growth path

For any growing variable $x(t)$, write

$$x(t) = e^{\gamma_x t} \tilde{x}(t), \quad (73)$$

and use a bar for its constant value on the balanced growth path (BGP). From free entry,

$$\gamma_L = \gamma_M + \gamma_F. \quad (74)$$

For the right-hand side of (59) to remain constant along the BGP, $\gamma_F + \eta\gamma_M = 0$. For the empirically relevant case $0 < \eta < 1$, this implies

$$\gamma_M = \frac{\gamma_L}{1 - \eta}, \quad \gamma_F = -\frac{\eta}{1 - \eta}\gamma_L. \quad (75)$$

The stationary skill distribution implies $\gamma_L = \gamma_N$. Consequently,

$$\gamma_M = \frac{\gamma_N}{1 - \eta}, \quad \gamma_F = -\frac{\eta}{1 - \eta}\gamma_N. \quad (76)$$

Substituting $F_g(t) = e^{\gamma_F t} \tilde{F}_g(t)$ and $L_g(t) = e^{\gamma_N t} \tilde{L}_g(t)$ into (60) gives

$$\frac{\dot{\tilde{F}}_g(t)}{\tilde{F}_g(t)} = \log \left[\frac{F_E \varepsilon^\eta \tilde{F}_g(t)^{\eta-1}}{c_L^\eta \tilde{L}_g(t)^\eta} \right] - \gamma_F. \quad (77)$$

At the BGP,

$$\bar{F}_g = [F_E \varepsilon^\eta (c_L \bar{L}_g)^{-\eta} e^{-\gamma_F}]^{1/(1-\eta)}, \quad (78)$$

$$\bar{M}_g = \frac{c_L \bar{L}_g}{\varepsilon \bar{F}_g} = \left[\frac{e^{\gamma_F} c_L \bar{L}_g}{F_E \varepsilon} \right]^{1/(1-\eta)}. \quad (79)$$

Because amenities satisfy $D_g(t) = D_g N_g(t)^{-\phi}$, equation (56) implies

$$\gamma_U = \frac{\alpha}{\varepsilon - 1} \gamma_M - [\theta(1 - \alpha) + \phi] \gamma_N = \left[\frac{\alpha}{(\varepsilon - 1)(1 - \eta)} - \theta(1 - \alpha) - \phi \right] \gamma_N. \quad (80)$$

The value function grows at the same rate, so $\gamma_V = \gamma_U$. Detrending (68) therefore gives

$$\begin{aligned} (\delta + \rho - \gamma_V) \tilde{V}_g(z, t) &= \tilde{U}_g(z, t) + \partial_t \tilde{V}_g(z, t) \\ &\quad + \lambda \sum_{d \in G} m_{gd}(z, t) [\delta_{gd}(z, t) \tilde{V}_d(z, t) - \tilde{V}_g(z, t)]. \end{aligned} \quad (81)$$

Migration shares are unaffected by detrending because the common factor $e^{\gamma_V t}$ cancels from (63).

The density is stationary in growth units. Defining

$$\tilde{\gamma}_N(t) \equiv \frac{\dot{\tilde{N}}(t)}{\tilde{N}(t)}, \quad (82)$$

the detrended Kolmogorov equation is

$$\begin{aligned} \partial_t \tilde{\psi}_g(z, t) = & \frac{\tilde{E}_g(t)}{\tilde{N}(t)} f(z) - (\delta + \gamma_N + \tilde{\gamma}_N(t)) \tilde{\psi}_g(z, t) \\ & - \lambda \left[(1 - m_{gg}(z, t)) \tilde{\psi}_g(z, t) - \sum_{i \neq g} m_{ig}(z, t) \tilde{\psi}_i(z, t) \right], \end{aligned} \quad (83)$$

where

$$\tilde{\gamma}_N(t) = \frac{\sum_g \tilde{E}_g(t)}{\tilde{N}(t)} - \delta - \gamma_N. \quad (84)$$

Along the BGP, all detrended variables are constant. Equations (81) and (83) become

$$(\delta + \rho - \gamma_V) \bar{V}_g(z) = \bar{U}_g(z) + \lambda \sum_{d \in G} \bar{m}_{gd}(z) [\bar{\delta}_{gd}(z) \bar{V}_d(z) - \bar{V}_g(z)], \quad (85)$$

$$0 = \frac{\bar{E}_g}{\bar{N}} f(z) - (\delta + \gamma_N) \bar{\psi}_g(z) - \lambda (1 - \bar{m}_{gg}(z)) \bar{\psi}_g(z) + \lambda \sum_{i \neq g} \bar{m}_{ig}(z) \bar{\psi}_i(z). \quad (86)$$

Finally, integrating (86) over skill and summing across regions eliminates gross migration flows and gives

$$\gamma_N = \frac{\sum_{g \in G} \bar{E}_g}{\bar{N}} - \delta. \quad (87)$$

Equations (54), (53), (78)–(86), together with (43) and (71), characterize the BGP. The numerical appendix describes their computation and the transition algorithm.

B Numerical Appendix

This appendix describes the numerical solution of the balanced growth path (BGP) and transition dynamics. The implementation uses sparse linear algebra and fixed-point iteration. All quantities in the transition algorithm are expressed in units detrended by the growth rates of the terminal BGP.

B.1 Discretization and notation

Let z_i , $i = 1, \dots, I$, denote the raw-skill grid and let ω_i denote the associated quadrature weights, chosen so that $\sum_i \omega_i f(z_i) = 1$. Define

$$a_{i,g} = a_g(z_i) = z_i \exp\{\chi_g(\log z_i - \mu_z)\}. \quad (88)$$

For any function $x_g(z)$, stack the values $x_{i,g} = x_g(z_i)$ in a vector $\mathbf{x} \in \mathbb{R}^{IG}$. The formulas below do not depend on whether the implementation orders the vector first by skill or first by region, provided all sparse operators use the same ordering.

It is useful to define

$$c_L \equiv \alpha + \theta(1 - \alpha), \quad \Gamma_U = \left(\frac{\varepsilon}{\varepsilon - 1}\right)^\alpha ((1 - \alpha)\theta)^{\theta(1-\alpha)}, \quad \Gamma_g = A_g^\alpha T_g^{\theta(1-\alpha)}. \quad (89)$$

For a positive value vector $\mathbf{v}$, migration shares at node i are

$$m_{i,gd}(\mathbf{v}) = \frac{(\kappa_{gd} v_{i,d})^\nu}{\sum_{r \in G} (\kappa_{gr} v_{i,r})^\nu}, \quad (90)$$

and the corresponding adjustment term is

$$\delta_{i,gd}(\mathbf{v}) = \frac{\kappa_{gd}}{G} m_{i,gd}(\mathbf{v})^{-(1+\nu)/\nu}. \quad (91)$$

Given shares $\mathbf{m}$ and adjustments $\boldsymbol{\delta}$, define the sparse HJB operator $\mathcal{A}(\mathbf{m}, \boldsymbol{\delta})$ by

$$[\mathcal{A}(\mathbf{m}, \boldsymbol{\delta})\mathbf{x}]_{i,g} = \lambda \left[\sum_{d \in G} m_{i,gd} \delta_{i,gd} x_{i,d} - x_{i,g} \right]. \quad (92)$$

For the forward equation, define the $G \times G$ migration generator at skill node i as

$$Q_i(g, d) = \begin{cases} \lambda m_{i,gd}, & d \neq g, \\ -\lambda(1 - m_{i,gg}), & d = g. \end{cases} \quad (93)$$

Every row of Q_i sums to zero. Let $\mathcal{Q}(\mathbf{m})$ be the block-diagonal matrix containing the Q_i blocks. The forward operator acting on the density is $\mathcal{Q}(\mathbf{m})^\top$.

B.2 Balanced growth path

B.2.1 Growth rates and static objects

Normalize detrended aggregate population to $\bar{N} = N_0$. The BGP population growth rate is

$$\gamma_N = \frac{\sum_{g \in G} \bar{E}_g}{N_0} - \delta. \quad (94)$$

Because effective labor per worker has a stationary distribution, $\gamma_L = \gamma_N$. For $0 < \eta < 1$, the remaining growth rates are

$$\gamma_M = \frac{\gamma_N}{1 - \eta}, \quad \gamma_F = -\frac{\eta}{1 - \eta} \gamma_N, \quad \gamma_V = \left[\frac{\alpha}{(\varepsilon - 1)(1 - \eta)} - \theta(1 - \alpha) - \phi \right] \gamma_N. \quad (95)$$

Given a candidate labor allocation $\bar{L}_g$, the BGP fixed cost and mass of firms are

$$\bar{F}_g = [F_E \varepsilon^\eta (c_L \bar{L}_g)^{-\eta} e^{-\gamma_F}]^{1/(1-\eta)}, \quad (96)$$

$$\bar{M}_g = \frac{c_L \bar{L}_g}{\varepsilon \bar{F}_g}. \quad (97)$$

For candidate wages $\bar{W}_g$, trade shares are

$$\pi_{gd} = \frac{\left[\tau_{gd} \bar{M}_g^{1/(1-\varepsilon)} \bar{W}_g / A_g \right]^{1-\sigma}}{\sum_{r \in G} \left[\tau_{rd} \bar{M}_r^{1/(1-\varepsilon)} \bar{W}_r / A_r \right]^{1-\sigma}}. \quad (98)$$

The flow-utility vector is then

$$u_{i,g} = D_g \bar{N}_g^{-\phi} a_{i,g} \frac{\Gamma_g}{\Gamma_U} \pi_{gg}^{\alpha/(1-\sigma)} \bar{M}_g^{\alpha/(\varepsilon-1)} \bar{L}_g^{-\theta(1-\alpha)}. \quad (99)$$

B.2.2 HJB solution

At the BGP, the discretized HJB system is

$$(\delta + \rho - \gamma_V) \mathbf{v} = \mathbf{u} + \mathcal{A}(\mathbf{m}(\mathbf{v}), \boldsymbol{\delta}(\mathbf{v})) \mathbf{v}. \quad (100)$$

The nonlinearity comes entirely from the migration shares. Introduce a fictitious time step $\Delta s > 0$. Given an iterate $\mathbf{v}^k$, compute $\mathbf{m}^k = \mathbf{m}(\mathbf{v}^k)$ and $\boldsymbol{\delta}^k = \boldsymbol{\delta}(\mathbf{v}^k)$, and solve

$$\left[\left(\frac{1}{\Delta s} + \delta + \rho - \gamma_V \right) \mathbf{I} - \mathcal{A}(\mathbf{m}^k, \boldsymbol{\delta}^k) \right] \mathbf{v}^{k+1} = \mathbf{u} + \frac{1}{\Delta s} \mathbf{v}^k. \quad (101)$$

Iteration continues until both the value update and the residual of (100) are below tolerance. A positive initial guess is

$$v_{i,g}^0 = \frac{u_{i,g}}{\delta + \rho - \gamma_V}. \quad (102)$$

B.2.3 Stationary forward equation

Let

$$e_{i,g} = \frac{\bar{E}_g}{N_0} f(z_i) \quad (103)$$

denote the entry density. Conditional on the migration shares obtained from the HJB, the stationary Kolmogorov forward equation is the sparse linear system

$$\left[(\delta + \gamma_N) \mathbf{I} - \mathcal{Q}(\mathbf{m})^\top \right] \boldsymbol{\psi} = \mathbf{e}. \quad (104)$$

Regional population and labor are recovered using the same quadrature rule as the skill density:

$$\hat{N}_g = N_0 \sum_{i=1}^I \omega_i \psi_{i,g}, \quad (105)$$

$$\hat{L}_g = N_0 \sum_{i=1}^I \omega_i a_{i,g} \psi_{i,g}. \quad (106)$$

The discretization should satisfy

$$\sum_{g \in G} \sum_{i=1}^I \omega_i \psi_{i,g} = 1 \quad (107)$$

up to the linear-solver and quadrature tolerances.

B.2.4 Outer BGP fixed point

The BGP solver takes the model parameters, regional fundamentals, bilateral trade and mobility costs, and the entrant density $f(z_i)$ as inputs. The procedure is:

1. Compute the growth rates in (94)–(95). Initialize positive vectors $\mathbf{W}^0$, $\mathbf{L}^0$, and $\mathbf{N}^0$, and normalize one wage, for example $W_1^0 = 1$.
2. At outer iteration k , use $\mathbf{L}^k$ in (96)–(97). Use $\mathbf{W}^k$, $\mathbf{L}^k$, $\mathbf{N}^k$, and $\mathbf{M}^k$ to compute trade shares and flow utility from (98)–(99).
3. Solve the nonlinear HJB using (101), and recover migration shares from (90).
4. Solve (104), then compute $\hat{\mathbf{N}}^k$ and $\hat{\mathbf{L}}^k$ from (105)–(106).

5. Evaluate (96)–(97) at $\widehat{\mathbf{L}}^k$. Solve for $\widehat{\mathbf{W}}^k$ from

$$\widehat{W}_g^k \widehat{L}_g^k = \sum_{d \in G} \pi_{gd} (\widehat{\mathbf{W}}^k, \widehat{\mathbf{M}}^k) \widehat{W}_d^k \widehat{L}_d^k, \quad g \in G, \quad (108)$$

together with the wage normalization.

6. Update the outer-loop variables with damping parameter $\zeta \in (0, 1]$:

$$\mathbf{x}^{k+1} = (1 - \zeta) \mathbf{x}^k + \zeta \widehat{\mathbf{x}}^k, \quad \mathbf{x} \in \{\mathbf{W}, \mathbf{L}, \mathbf{N}\}. \quad (109)$$

Stop when the changes in all three vectors and the residuals of trade balance, the HJB, and the forward equation are below tolerance.

B.3 Transition dynamics

B.3.1 Time grid and boundary conditions

Let $t_n = n\Delta t$, $n = 0, \dots, N_T$. Superscripts 1 and 2 denote the initial and terminal BGPs. The transition uses

$$\boldsymbol{\psi}^0 = \bar{\boldsymbol{\psi}}^1, \quad \mathbf{v}^{N_T} = \bar{\mathbf{v}}^2, \quad \widetilde{F}_g^0 = \bar{F}_g^1. \quad (110)$$

The horizon is chosen so that the last simulated allocations are close to the terminal BGP. Transition variables are detrended using the terminal growth rates γ_N^2 , γ_F^2 , and γ_V^2 .

Aggregate detrended population follows

$$\dot{\widetilde{N}}(t) = \sum_{g \in G} \widetilde{E}_g(t) - (\delta + \gamma_N^2) \widetilde{N}(t). \quad (111)$$

A forward Euler step is

$$\widetilde{N}^{n+1} = \widetilde{N}^n + \Delta t \left[\sum_g \widetilde{E}_g^n - (\delta + \gamma_N^2) \widetilde{N}^n \right], \quad (112)$$

and

$$\widetilde{\gamma}_N^n = \frac{\sum_g \widetilde{E}_g^n}{\widetilde{N}^n} - \delta - \gamma_N^2. \quad (113)$$

B.3.2 Fixed costs, firms, trade, and utility

Given a candidate labor path, the detrended fixed cost obeys

$$\frac{\dot{\widetilde{F}}_g(t)}{\widetilde{F}_g(t)} = \log \left[\frac{F_E \varepsilon^\eta \widetilde{F}_g(t)^{\eta-1}}{c_L^\eta \widetilde{L}_g(t)^\eta} \right] - \gamma_F^2. \quad (114)$$

To preserve positivity, the implementation advances $x_g^n = \log \tilde{F}_g^n$:

$$x_g^{n+1} = x_g^n + \Delta t \left\{ \log F_E + \eta \log \left(\frac{\varepsilon}{c_L} \right) + (\eta - 1)x_g^n - \eta \log \tilde{L}_g^n - \gamma_F^2 \right\}, \quad \tilde{F}_g^{n+1} = e^{x_g^{n+1}}. \quad (115)$$

The mass of firms is

$$\tilde{M}_g^n = \frac{c_L \tilde{L}_g^n}{\varepsilon \tilde{F}_g^n}. \quad (116)$$

At each date, compute trade shares from

$$\pi_{gd}^n = \frac{\left[\tau_{gd} (\tilde{M}_g^n)^{1/(1-\varepsilon)} \tilde{W}_g^n / A_g \right]^{1-\sigma}}{\sum_{r \in G} \left[\tau_{rd} (\tilde{M}_r^n)^{1/(1-\varepsilon)} \tilde{W}_r^n / A_r \right]^{1-\sigma}} \quad (117)$$

and flow utility from

$$u_{i,g}^n = D_g (\tilde{N}_g^n)^{-\phi} a_{i,g} \frac{\Gamma_g}{\Gamma_U} (\pi_{gg}^n)^{\alpha/(1-\sigma)} (\tilde{M}_g^n)^{\alpha/(\varepsilon-1)} (\tilde{L}_g^n)^{-\theta(1-\alpha)}. \quad (118)$$

B.3.3 Backward solution of the HJB

At time t_n , approximate $\partial_t \tilde{V}$ by

$$\partial_t \tilde{V}_g(z_i, t_n) \approx \frac{v_{i,g}^{n+1} - v_{i,g}^n}{\Delta t}. \quad (119)$$

Conditional on migration shares, the implicit system for $\mathbf{v}^n$ is

$$\left[\left(\delta + \rho - \gamma_V^2 + \frac{1}{\Delta t} \right) \mathbf{I} - \mathcal{A}(\mathbf{m}^n, \delta^n) \right] \mathbf{v}^n = \mathbf{u}^n + \frac{1}{\Delta t} \mathbf{v}^{n+1}. \quad (120)$$

Since $\mathbf{m}^n$ depends on $\mathbf{v}^n$, solve (120) by an inner fixed point at each date. Starting from $\mathbf{v}^{n,0} = \mathbf{v}^{n+1}$, compute shares and adjustments from $\mathbf{v}^{n,k}$, solve the resulting linear system for $\mathbf{v}^{n,k+1}$, and iterate to convergence. Apply this procedure backward for $n = N_T - 1, \dots, 0$.

B.3.4 Forward solution of the distribution

Let $\mathbf{e}^n$ have elements

$$e_{i,g}^n = \frac{\tilde{E}_g^n}{\tilde{N}_g^n} f(z_i). \quad (121)$$

An implicit step for the detrended forward equation is

$$\frac{\boldsymbol{\psi}^{n+1} - \boldsymbol{\psi}^n}{\Delta t} = \mathbf{e}^n + \left[\mathcal{Q}(\mathbf{m}^n)^\top - (\delta + \gamma_N^2 + \tilde{\gamma}_N^n) \mathbf{I} \right] \boldsymbol{\psi}^{n+1}. \quad (122)$$

Equivalently,

$$\boldsymbol{\psi}^{n+1} = \left\{ \mathbf{I} - \Delta t \left[\mathcal{Q}(\mathbf{m}^n)^\top - (\delta + \gamma_N^2 + \tilde{\gamma}_N^n) \mathbf{I} \right] \right\}^{-1} (\boldsymbol{\psi}^n + \Delta t \mathbf{e}^n). \quad (123)$$

Starting from $\bar{\boldsymbol{\psi}}^1$, iterate (123) forward for $n = 0, \dots, N_T - 1$.

Regional allocations are recovered from

$$\hat{N}_g^n = \tilde{N}^n \sum_{i=1}^I \omega_i \psi_{i,g}^n, \quad (124)$$

$$\hat{L}_g^n = \tilde{N}^n \sum_{i=1}^I \omega_i a_{i,g} \psi_{i,g}^n. \quad (125)$$

B.3.5 Transition-path fixed point

The complete algorithm is:

1. Solve the BGP under the initial parameters to obtain $\{\bar{\mathbf{v}}^1, \bar{\boldsymbol{\psi}}^1, \bar{\mathbf{F}}^1, \bar{\mathbf{M}}^1\}$, and solve the BGP under the terminal parameters to obtain the corresponding objects with superscript 2.
2. Choose N_T and Δt . Compute the aggregate population path from (112). Initialize paths for regional labor, population, and wages, using smooth interpolations between the two BGPs. Normalize the same wage at every date.
3. At path iteration k :
 - (a) Given $\{\tilde{\mathbf{L}}^{n,k}\}_{n=0}^{N_T}$, compute the fixed-cost path forward from (115) and the mass of firms from (116).
 - (b) Compute trade shares and flow utility at every date using (117)–(118).
 - (c) Solve the HJB backward from $\bar{\mathbf{v}}^2$ using (120). Store the implied migration shares at every date.
 - (d) Solve the forward equation from $\bar{\boldsymbol{\psi}}^1$ using (123). Recover $\hat{\mathbf{N}}^{n,k}$ and $\hat{\mathbf{L}}^{n,k}$ from (124)–(125).
 - (e) For each date, recompute fixed costs and firms using the updated labor path, and solve the balanced-trade system

$$\hat{W}_g^{n,k} \hat{L}_g^{n,k} = \sum_{d \in G} \pi_{gd} (\hat{\mathbf{W}}^{n,k}, \hat{\mathbf{M}}^{n,k}) \hat{W}_d^{n,k} \hat{L}_d^{n,k} \quad (126)$$

subject to the wage normalization.

- (f) Update the entire paths for $\mathbf{W}$, $\mathbf{L}$, and $\mathbf{N}$ using damping as in (109). Stop when the maximum proportional change in every path and all equilibrium residuals are below tolerance.

C Estimation Appendix

TABLE 5 REGRESSION OF PRICES ON DISTANCE

	$\log p_{i,m,n,j,t}$			
	(1)	(2)	(3)	(4)
ζ_τ	0.283*** (0.001)	0.169*** (0.001)	0.283*** (0.001)	0.168*** (0.001)
Same State	-0.467*** (0.002)	-0.210*** (0.002)	-0.467*** (0.002)	-0.211*** (0.002)
Orig. State	✓			
Dest. State	✓			
Orig. State * Prod.		✓		
Dest. State * Prod.		✓		
Orig. State * Quarter			✓	
Dest. State * Quarter			✓	
Orig. State * Prod. * Quarter				✓
Dest. State * Prod. * Quarter				✓
N	10,526,115	10,526,114	10,526,115	10,525,995
R^2	0.114	0.396	0.115	0.400